Sensitivity Analysis and Robust Estimation in Additive IV

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Abstract

Recent work shows that in additive instrumental variables (IV) analysis, the robustness to some small failure of the identifying conditional moment restriction depends on the strength of the a priori smoothness or functional-form assumptions on the structural relationship between outcomes and treatments. In this work we develop methods for empirical sensitivity analysis that can help applied researchers assess the terms of this trade-off. We suggest a related estimator targeting the midpoint of an identified interval, which minimizes worst-case scalar error under a relaxation of the point-identifying assumptions. In effect, this estimator applies a novel regularization scheme, one that is motivated by the possibility that the IV moment condition is misspecified. We apply our methods to the consumer demand data previously analyzed in Blundell et al. (2007) and Horowitz (2011).

Instrumental variables (IV) analysis provides a means of empirically evaluating causal effects when treatment is endogenous. However, the validity of this approach rests on strong assumptions: the instruments must be assigned as if by a randomized experiment and they must not directly impact outcomes. Additive IV models (e.g., linear IV or the nonparametric IV model of Newey and Powell (2003) and Ai and Chen

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(2003)) also require that the outcome model has a particular additive structure, which may be difficult to motivate using setting-specific knowledge or economic theory. As a result, applied researchers typically argue not that their IV model is exactly correctly specified, but rather that it holds to a close approximation.[1]

Even some small degree of misspecification may be consequential. Prior work shows that in additive IV models in particular, statistical inference can be highly misleading under misspecification, but this depends crucially on some empirically estimable quantities and on certain modeling choices made by the researcher. In particular, Deaner (2025) demonstrates that the sensitivity of IV inference to misspecification depends crucially on the object of interest (e.g., a weighted average partial effect) and the flexibility of the additive IV model considered, with more flexible models associated with greater sensitivity. Additionally, the classic work of Bound et al. (1995) shows that the bias in linear IV estimates resulting from misspecification is amplified if the instruments are weakly correlated with treatments.

Motivated by these concerns, the present work provides methods for empirical sensitivity analysis in flexible additive IV models, particularly models in which the causal relationship between treatments and outcomes (the 'structural function') is not assumed to take any particular functional form. Our approach also yields a related point estimator, which is 'robust' in the sense that it targets an estimand with minimal worst-case scalar error under misspecification. We show that the robustness of the estimator acts as a form of 'regularization' (see e.g., Chen and Pouzo, 2012) and derive conditions for consistency under correct specification as the misspecification tolerance shrinks.

We quantify sensitivity to misspecification similarly to Deaner (2025), which in turn follows the conceptual approach of Masten and Poirier (2018). Methods for sensitivity analysis are increasingly popular in applied economics and include local methods (e.g., Andrews et al. (2017), Armstrong and Kolesar (2018), Bonhomme and Weidner (2022)), and global methods (e.g., Altonji et al. (2005), Masten and Poirier (2018), Oster (2019)). The present work takes a global approach. To be more precise, we assess sensitivity by estimating the identified set when the point-identifying assumptions are relaxed. Additive IV models are point-identified by a conditional moment equality. We relax this conditional moment equality, with the degree of relaxation controlled by a parameter b. Larger values of b allow for more

[1] See Conley et al. (2008) for further discussion.

misspecification, and $b = 0$ corresponds to correct specification. Under this relaxation, objects of interest are (in general) only partially identified. We say that an empirical finding is robust to misspecification if it holds up when one allows for some small degree of misspecification, that is, if it is implied by the identified set for some small but non-zero b. For example, if the identified set for an average partial effect (APE) contains only positive values, then the finding that the APE is positive is robust to misspecification.

In practice, the identified set is not known a priori, and must be estimated empirically: we provide methods to do so. The mid-point of the identified set for an object of interest is, by construction, an estimand with the lowest worst-case bias under the assumptions that define the set. Our robust estimator is an estimate of this mid-point. Both the set estimation and robust estimator can be evaluated by linear programming at little computational cost. This approach also allows researchers to easily impose additional linear restrictions, like smoothness or shape restrictions, without complicating estimation.

We show that the set estimation problem is 'well-posed' in a certain sense. Practically speaking, this means that the identified set can be estimated at a rate that does not depend on any 'sieve measure of ill-posedness' or 'sieve modulus of continuity' (Chen and Pouzo, 2012). This is because the relaxation of the conditional moment restriction effectively acts to regularize the estimation problem. This property is inherited by the robust estimator. The discrepancy b specified by the researcher acts to control the degree of regularization. Unlike standard methods that may use the sieve dimension, Tikhonov regularization parameter, or quasi-Bayes prior (see Kankanala, 2025), b indexes an interpretable family of departures that the researcher is unwilling to rule out, and we provide guidance for calibrating this quantity.

We apply our methods to the empirical setting shared by Blundell et al. (2007) and Horowitz (2011) who use nonparametric IV methods to estimate shape-invariant Engel curves using data from The British Family Expenditure Survey. We argue that in this setting instruments are unlikely to be fully exogenous, in which case the NPIV moment restriction will not hold exactly. We use our methodology to assess which features of the structural Engel curve for food can be inferred robustly.

Finally, we examine the performance of our robust point estimator under correct specification. We demonstrate that it performs comparably with (neither uniformly

better nor worse than) state-of-the-art NPIV point estimators even under correct specification.

1 Modeling Misspecification in Additive IV

Let Y be an outcome, X a treatment or vector of treatments, and Z a vector of instrumental variables (IVs). Let S_X and S_Z denote their respective treatment and instrument supports. We consider models of the following form

$$Y = h_0(X) + \varepsilon. \tag{1.1}$$

The function h_0 is understood to capture a causal relationship between the outcome Y and treatment/s X. As a leading example, let $Y(x)$ be the 'potential outcome' under a fixed counterfactual treatment level x and define $h_0(x) := E[Y(x)]$, then $h_0(x)$ is the average potential outcome from treatment level x. Interest is in either h_0 itself or perhaps more commonly, some linear functional of h_0, for example the average causal effect of a 10% increase in the treatment level, which is, in the case of $h_0(x)$ the average potential outcome, $E[h_0(1.1X) - h_0(X) \mid X \in S_*]$, where $S_* = \{x \in S_X : 1.1x \in S_X\}$ and $P(X \in S_*) > 0$. This restricts the comparison to treatment values with common support.

The corresponding additive IV model is obtained by imposing the restriction that the additive 'structural residual' ε is conditionally mean zero given the instrument/s Z, that is,

$$E[Y - h_0(X) \mid Z] = 0, \tag{1.2}$$

where the equality should be understood to hold with probability 1 (we follow this convention throughout the paper). Under a condition known as 'statistical completeness', the conditional moment restriction above identifies h_0, even in the absence of any further restrictions on this object (Newey and Powell, 2003; Ai and Chen, 2003). If the conditional moment restriction (1.2) holds then we say the model is 'correctly specified'. We do not consider the model correctly specified if the restriction holds for *some* h_0, only if it holds for the particular h_0 of interest (e.g., the average potential outcome function). The desire to estimate a particular causally-interpretable object is what motivates the use of IV methods over say, regression of Y on X.

Deaner (2025) provides a causal interpretation of the model above (see Appendix A.1 in that paper for details). He shows that when $h_0(x)$ is an average potential outcome from counterfactual treatment level x, the conditional moment restriction above follows from instrumental validity restrictions similar to those of Angrist et al. (1996), namely an exclusion restriction and random assignment of the instrument. However, unlike in Angrist et al. (1996), the additive structure in the model above necessitates a strong additional assumption 'exogenous effects', which states that any heterogeneity in treatment effects is jointly mean independent of both the instruments and treatments. Deaner (2025) shows that if any one of these conditions fails, then the model is generally misspecified in the sense that (1.2) does not hold.

In order to examine sensitivity to misspecification Deaner (2025) suggests one relax the conditional moment equality. Define the function u_0 by

$$u_0(Z) := E[Y - h_0(X) \mid Z].$$

Using this notation, correct specification is $u_0 = 0$. As a leading example, Deaner (2025) considers the conditional moment *inequality*:

$$|u_0(Z)| \leq b, \tag{1.3}$$

where the inequality should be understood to hold with probability 1.[2] In this leading case, the positive scalar b bounds the magnitude of misspecification, with $b = 0$ corresponding to correct specification. A small but strictly positive b means the specification is approximately correct: the conditional moment equality (1.2) may not hold, but it must *almost hold* in that $E[Y - h_0(X) \mid Z]$ must be uniformly close to zero.

The inequality (1.3) provides a means of modeling misspecification that is agnostic about the precise source of that misspecification, e.g., violation of exogenous effects or of the exclusion restriction. Deaner (2025) characterizes u_0 under various sources of misspecification (e.g., violation of the exclusion restriction) and provides corresponding interpretations of the bound b. As a leading example, suppose that the IV exclusion restriction fails so that there is an additive direct effect of the instrument

[2]The analysis in Deaner (2025) is somewhat more general. He imposes a bound b on the norm of u_0 and allows for general choices of this norm. The restriction $|u_0(Z)| \leq b$ corresponds to a bound on the essential supremum norm and is thus a special case. Deaner (2025) also considers additional a priori restrictions on u_0, for example, that $E[u_0(Z)] = 0$.

Z on the outcome Y. Let γ_0 capture this direct effect so that the potential outcome from counterfactual treatment level x and IV level z is

$$Y(x, z) = h_0(x) + \gamma_0(z) + \varepsilon.$$

Normalize $E[\gamma_0(Z)] = 0$ and suppose the instrument is randomly assigned so that $E[\varepsilon|Z] = 0$. Then we have $u_0 = \gamma_0$. The average direct effect of setting the instrument to level z is then $E[Y(x, z) - Y(x, Z)] = \gamma_0(z)$, and so (1.3) imposes that the average direct effect has magnitude smaller than b.

We allow for more general relaxations than (1.3). Firstly, researchers may wish to impose the normalization $E[u_0(Z)] = 0$, as discussed in Deaner (2025). Our analysis allows for such normalizations. Additionally, in place of (1.3), we allow researchers to restrict u_0 to lie in a set $\mathcal{U}_b$ which is parameterized by $b \geq 0$. Formally,

$$u_0 \in \mathcal{U}_b. \tag{1.4}$$

We allow for various choices of $\mathcal{U}_b$ and the restriction (1.3) arises as a special case. We assume that $\mathcal{U}_b$ contains zero (more precisely, a function that is identically zero) and so the restriction $u_0 \in \mathcal{U}_b$ does not preclude correct specification. Additionally, the sets $\mathcal{U}_b$ must be nested in b, that is, for $b' > b$ we have $\mathcal{U}_b \subseteq \mathcal{U}_{b'}$, so larger values permit more departures from the conditional moment equality. We define the sets $\mathcal{U}_b$ by linear inequality restrictions in Section 1.2. The parameter b can bound levels, derivatives, or other specified features of the departure. Its units and interpretation depend on the chosen restrictions. For example, suppose Z is scalar with support $[0, 1]$, after a normalization of units. We allow one to instead restrict u_0 so that

$$|u_0'(Z)| \leq b \tag{1.5}$$

where u_0' is the derivative of u_0 and its absolute value is understood to be infinite at points of non-differentiability. Combined with a normalization $E[u_0(Z)] = 0$, setting $b = 0$ corresponds to correct specification of the conditional moment equality. In the earlier example of an additive direct effect γ_0, this restricts the magnitude of the average partial effect of the instrument to be at most b:

$$\left| E\left[\frac{\partial}{\partial z} Y(x, z)\right] \right| = |\gamma_0'(z)| \leq b.$$

1.1 An Approach to Sensitivity Analysis

The relaxed condition (1.4) is central to our approach to sensitivity analysis. An inference made under the assumption (1.2) is 'robust to misspecification' if it remains valid under the relaxation (1.4) for some small $b > 0$. That is, one does not need to believe the conditional moment restriction holds exactly for the finding to be valid and instead it suffices that the condition only holds approximately. If it remains valid under (1.4) for a larger b then it is relatively more robust.

To formalize this, consider that, under statistical completeness, the moment equality (1.2) point identifies h_0, whereas (1.4) for a fixed $b > 0$ generally only partially identifies h_0. The corresponding identified set captures everything that could be learned about h_0 from the data under the assumption (1.4). If say, every element of the identified set is monotone, then the finding that h_0 is monotone is robust. Under this approach, empirical sensitivity analysis amounts to the problem of estimating (and performing inference on) the identified set. This may be the identified set for h_0 or for a particular functional of interest. Note that this approach to sensitivity analysis is estimator-free: rather than assess say, the bias of a particular IV estimator under misspecification, we directly assess the robustness of conclusions one might draw from the data.

We now formally define the identified set. Let $\mathcal{H}$ be a set of functions assumed to contain h_0. This set may incorporate additional a priori restrictions on h_0, for example, if the researcher believes h_0 is convex then one can impose this assumption by letting $\mathcal{H}$ contain only convex functions. The identified set, given the assumptions that $h_0 \in \mathcal{H}$ and (1.4) holds, is then formally defined by

$$\Theta_b := \left\{ h \in \mathcal{H} : \; E[Y - h(X) \mid Z = \cdot] \in \mathcal{U}_b \right\}. \tag{1.6}$$

The identified set varies with the choice of b which indexes the allowed departures. A larger value of b entails a larger identified set and thus under a larger b less can be learned from the data about h_0. By varying b and inspecting the identified set (or in practice, an estimate of the set), one can assess which features of h_0 can be inferred under relaxations of the conditional moment restriction.

In practice, researchers may not be interested in the whole structural function h_0 but a functional thereof. Recall the example in the previous subsection in which interest is in the average causal effect of a 10% increase in the treatment level

$E[h_0(1.1X) - h_0(X) \mid X \in S_*]$. This object is a scalar functional of h_0, it is a linear functional. Our empirical methods accommodate linear functionals. Another important linear functional is the value of h_0 at a fixed point x, which is simply $h_0(x)$. Let $\mathbb{L}$ be a scalar functional defined on $\mathcal{H}$. That is, $\mathbb{L}[h]$ takes the function $h \in \mathcal{H}$ and returns a scalar. The identified set for $\mathbb{L}[h_0]$ is the image of Θ_b under $\mathbb{L}$, denoted $\mathbb{L}[\Theta_b]$, and it can be defined directly by

$$\mathbb{L}[\Theta_b] := \left\{ \mathbb{L}[h] : \ h \in \mathcal{H} \ \text{ and } \ E[Y - h(X) \mid Z = \cdot] \in \mathcal{U}_b \right\}. \tag{1.7}$$

For the leading supremum restriction, a key finding in Deaner (2025) is that the additional assumptions a researcher places on h_0 (embodied in the restrictiveness of $\mathcal{H}$) can be highly consequential for the sensitivity to misspecification. Stronger restrictions on h_0 correspond to lower sensitivity. More precisely, the rate at which the diameter of Θ_b shrinks as b approaches zero depends crucially on $\mathcal{H}$, and if only very weak restrictions are placed on $\mathcal{H}$ then the diameter of the identified set does not shrink to zero with b. This phenomenon may or may not also apply to $\mathbb{L}[\Theta_b]$, depending on the particular linear functional of interest $\mathbb{L}$ and the joint distribution of X and Z. In sum, there is a trade-off between the flexibility of the a priori restrictions placed on h_0 and the sensitivity to misspecification of the conditional moment equality. Estimation of Θ_b and $\mathbb{L}[\Theta_b]$ for different choices of $\mathcal{H}$ may aid researchers in making this trade-off.

1.2 A Tractable Characterization of the Identified Set

Let $\mathcal{X}$ be a linear space of bounded continuous real-valued functions on S_X. Let $S_H \subseteq \mathbb{R}^p$ index the shape restrictions and let $\mathbb{T}_H$ be a known linear operator from $\mathcal{X}$ to $\mathbb{R}^d$-valued functions on S_H. For a known vector function C, take

$$\mathcal{H} = \{h \in \mathcal{X} : \mathbb{T}_H[h](x) \leq C(x), \forall x \in S_H\}. \tag{1.8}$$

The inequality $\mathbb{T}_H[h](x) \leq C(x)$ (and other inequalities between vectors) should be understood to hold componentwise. Often $S_H = S_X$, after a normalization of the treatment support. For example, $0 \leq h \leq 1$ and $|h''| \leq c$ correspond to $\mathbb{T}_H[h] = (h, -h, h'', -h'')'$ and $C = (1, 0, c, c)'$ on a compact interval, with $\mathcal{X} = C^2(S_X)$.

Similarly, let $\mathcal{Z}$ be a linear space of bounded continuous functions on S_Z. Let $W_U \subseteq \mathbb{R}^{p_U}$ and let $\mathbb{T}_U$ be a known linear operator from $\mathcal{Z}$ to $\mathbb{R}^{d_U}$-valued functions on W_U. For each scalar relaxation index $b \geq 0$, write $b(w)$ for the known bound vector at w, using b for both the index and its associated function. Define

$$\mathcal{U}_b = \{u \in \mathcal{Z} : \mathbb{T}_U[u](w) \leq b(w), \forall w \in W_U\}. \tag{1.9}$$

The bounds $b(w)$ are componentwise nondecreasing in the scalar index b. For the leading example (1.3), take $W_U = S_Z$, $\mathbb{T}_U[u](w) = (u(w), -u(w))'$ and $b(w) = (b, b)'$. For the scalar derivative restriction (1.5), one can instead take $W_U = [0, 1]$ and

$$\mathcal{Z} = C^1([0, 1]), \qquad \mathbb{T}_U[u](w) = (u'(w), -u'(w))', \qquad b(w) = (b, b)'. \tag{1.10}$$

Given the characterizations of $\mathcal{H}$ and $\mathcal{U}_b$ above, the identified set consists of all functions $h \in \mathcal{X}$ that satisfy the constraints below

$$\mathbb{T}_U[E[Y - h(X) \mid Z = \cdot]](w) \leq b(w), \qquad \forall w \in W_U \tag{1.11}$$

$$\mathbb{T}_H[h](x) \leq C(x), \qquad \forall x \in S_H \tag{1.12}$$

Both $\mathcal{H}$ and $\mathcal{U}_b$ are convex, so Θ_b is convex. For a scalar linear functional $\mathbb{L}$ on $\mathcal{X}$, its image is an interval whenever the identified set is nonempty. The lower and upper end-points of the interval can then be written as:

$$\underline{\theta}_{\mathbb{L}}(b) = \inf_{h \in \mathcal{X}} \mathbb{L}[h] \quad \text{s.t. (1.11) and (1.12)} \tag{1.13}$$

$$\bar{\theta}_{\mathbb{L}}(b) = \sup_{h \in \mathcal{X}} \mathbb{L}[h] \quad \text{s.t. (1.11) and (1.12).} \tag{1.14}$$

These are linear programs: the objectives are linear, and the constraints are linear inequalities in h. Whenever discussing midpoints, we require a nonempty set and finite endpoints.

For point evaluation $\mathbb{L}[h] = h(x)$, write

$$\underline{\theta}(x; b) = \inf_{h \in \mathcal{X}} h(x) \quad \text{s.t. (1.11) and (1.12)} \tag{1.15}$$

$$\bar{\theta}(x; b) = \sup_{h \in \mathcal{X}} h(x) \quad \text{s.t. (1.11) and (1.12).} \tag{1.16}$$

These are the lower and upper envelopes of Θ_b. They display pointwise identified intervals but do not describe all joint restrictions on h: an arbitrary function between them need not belong to Θ_b. We suppress b when it is fixed.

We can report a mean-zero normalization of the residual by first bounding the centered structural function using the linear functional

$$\mathbb{L}_x[h] := h(x) - E[h(X)], \tag{1.17}$$

and then adding $E[Y]$, which is identified from the observable distribution. The resulting lower and upper bounds on the normalized version of the structural function are

$$E[Y] + \underline{\theta}_{\mathbb{L}_x}(b), \qquad E[Y] + \bar{\theta}_{\mathbb{L}_x}(b).$$

For a feasible h, write $u_h(Z) = E[Y - h(X) \mid Z]$. Replacing h by $h + E[Y] - E[h(X)]$ replaces its residual by $u_h - E[u_h(Z)]$, which has mean zero. These bounds coincide with imposing $E[Y - h(X)] = 0$ in the original program if this normalization preserves both $\mathcal{H}$ and $\mathcal{U}_b$. This holds, for example, when the shape restrictions are invariant to constant shifts and the residual restrictions concern only derivatives. With level restrictions, the two problems can differ; the displayed bounds concern the normalized target and require no equality constraint.

1.3 Robust Point Estimands

Our analysis motivates a robust point estimator of the structural function h_0 or a scalar linear functional $\mathbb{L}[h_0]$. Our approach is to estimate the midpoint of the corresponding identified interval. Define

$$\begin{aligned} \theta^{\text{mid}}_{\mathbb{L}}(b) &= \tfrac{1}{2}\{\underline{\theta}_{\mathbb{L}}(b) + \bar{\theta}_{\mathbb{L}}(b)\}, \\ \theta^{\text{mid}}(x; b) &= \tfrac{1}{2}\{\underline{\theta}(x; b) + \bar{\theta}(x; b)\}. \end{aligned} \tag{1.18}$$

An estimate of $\theta^{\text{mid}}_{\mathbb{L}}$ is our robust estimate of $\mathbb{L}[h_0]$, and an estimate of $\theta^{\text{mid}}(x)$ is our robust estimate of $h_0(x)$. The midpoint function $\theta^{\text{mid}}(\cdot)$ can be understood as a 'regularized' solution to the integral equation $E[Y - h(X)|Z] = 0$, where h is the unknown in the equation. The degree of regularization is controlled by b. We show this formally in Section 3.1. Regularization is crucial to good performance of point estimates in non-parametric additive IV models under correct specification (see e.g.,

Chen and Pouzo, 2012). The regularization here results directly from the definition of these estimates as midpoints of the identified set.

These targets are robust in the sense that they minimize worst-case squared scalar loss under the assumptions defining the identified set. This decision-theoretic motivation for estimating the midpoint of an identified interval is also studied by Song (2014) and Dominitz and F. Manski (2017). We first explain the oracle decision problem, then relate it to the minimax risk of feasible estimators. Throughout this discussion, the misspecification tolerance b is fixed.

Consider a deterministic estimate c of $\mathbb{L}[h_0]$, with squared loss $(c - \mathbb{L}[h_0])^2$. If all that is known about h_0 is that it lies in Θ_b, the worst possible loss is $\sup_{h\in\Theta_b}(c-\mathbb{L}[h])^2$. The unique minimizer is the midpoint $\theta_{\mathbb{L}}^{\text{mid}}$. Let $d_{\mathbb{L}} = \bar{\theta}_{\mathbb{L}} - \underline{\theta}_{\mathbb{L}}$, then the midpoint achieves the minimized loss

$$\inf_{c\in\mathbb{R}} \sup_{h\in\Theta_b} (c - \mathbb{L}[h])^2 = d_{\mathbb{L}}^2/4. \tag{1.19}$$

This is an oracle decision because the identified interval depends on the unknown distribution of observables.

For a formal statistical argument, let a model m consist of a joint distribution of (X, Z, ε) and a structural function h_m. Together with $Y = h_m(X)+\varepsilon$, these determine a distribution P_m of observables (X, Z, Y). Let $\mathcal{M}$ be the class of models satisfying the restrictions that define the identified set: $h_m \in \mathcal{H}$ and $E_{P_m}[Y - h_m(X) \mid Z = \cdot] \in \mathcal{U}_b$. Let $\mathcal{P}$ be a class of distributions of observables, and suppose the true model belongs to $\mathcal{M}$ and its observable distribution P_0 belongs to $\mathcal{P}$. Write $\Theta_b(P)$, $\theta_{\mathbb{L}}^{\text{mid}}(P)$, and $d_{\mathbb{L}}(P)$ to make the dependence on an observable distribution P explicit. The class $\mathcal{H}$ and the rule defining $\mathbb{L}$ are fixed. If the functional involves observable moments, they are evaluated under the model's observable law, with that dependence suppressed in the notation. In particular, the functional is the same throughout each observational-equivalence class.

Suppose $\hat{c}_n$ is a measurable estimator based on n independent observations from P_m. Its maximum expected squared loss over the models under consideration is

$$R_n(\hat{c}_n; \mathcal{P}) := \sup_{m\in\mathcal{M}:\, P_m\in\mathcal{P}} E_{P_m^{\otimes n}}[(\hat{c}_n - \mathbb{L}[h_m])^2]. \tag{1.20}$$

The subscript $P_m^{\otimes n}$ on the expectation indicates that this is the mean when the observations are independently and identically distributed according to the probability distribution P_m. Every estimator with finite maximum risk satisfies

$$\begin{aligned} R_n(\hat{c}_n; \mathcal{P}) &\geq \sup_{m \in \mathcal{M}\colon P_m = P_0} \left(E_{P_0^{\otimes n}}[\hat{c}_n] - \mathbb{L}[h_m]\right)^2 \\ &= \sup_{h \in \Theta_b(P_0)} \left(E_{P_0^{\otimes n}}[\hat{c}_n] - \mathbb{L}[h]\right)^2 \\ &\geq \inf_{c \in \mathbb{R}} \sup_{h \in \Theta_b(P_0)} (c - \mathbb{L}[h])^2 = \frac{d_{\mathbb{L}}(P_0)^2}{4}. \end{aligned} \tag{1.21}$$

The first inequality restricts attention to models with the same observable distribution P_0 and applies Jensen's inequality. These models are observationally equivalent, so the distribution of $\hat{c}_n$ is the same under all of them. The first equality holds because their structural functions are precisely the elements of $\Theta_b(P_0)$: for any h in that set, defining $\varepsilon = Y - h(X)$ under P_0 constructs a compatible model. The final equality is (1.19). The risk lower bound also holds trivially for estimators with infinite maximum risk.

The minimum in (1.21) is achieved by setting $c = \theta^{\text{mid}}_{\mathbb{L}}$. That is, $\theta^{\text{mid}}_{\mathbb{L}}$ is an oracle estimate of $\mathbb{L}[h_0]$ that achieves minimax risk under the assumptions that define the identified set. If the identified-interval widths are uniformly bounded over the compatible observable laws and $\hat{\theta}^{\text{mid}}_{\mathbb{L}}$ is a feasible estimator that converges in mean square to $\theta^{\text{mid}}_{\mathbb{L}}$ uniformly over $P \in \mathcal{P}$, then its maximum risk converges to the minimax risk. Similar reasoning applies for a uniformly mean-square consistent estimator of $\theta^{\text{mid}}(x)$.[3]

For the normalization in (1.17), first form the midpoint for the centered target and then add $E[Y]$. The resulting point estimand is $E[Y] + \theta^{\text{mid}}_{\mathbb{L}_x}(b)$. At a fixed observable law, this common shift leaves interval widths and the minimax squared-error calculation unchanged.

[3]The restriction of uniformity over $\mathcal{P}$ is stronger than needed, uniformity over the subset of distributions in $\mathcal{P}$ which are compatible with models in $\mathcal{M}$ would suffice.

2 Estimation

We provide methods to estimate $\underline{\theta}_{\mathbb{L}}$, $\bar{\theta}_{\mathbb{L}}$, $\underline{\theta}(x)$, and $\bar{\theta}(x)$, using sample data

$$\{(Y_i, X_i, Z_i)\}_{i=1}^n.$$

In order to construct feasible estimates, we replace the infeasible optimization problems in (1.13), (1.14), (1.15), and (1.16), with feasible finite-dimensional empirical analogs.

First, we replace the infinite-dimensional problem with a finite-dimensional one. The infeasible problems maximize or minimize over a set of functions $\mathcal{X}$ and we approximate this with a finite-dimensional sieve space. Second, we replace the conditional moment in the constraints (1.11) with an empirical estimate. Finally, we impose the constraints only on finite grids of points.

Let Φ_n be a length-K_n column vector of basis functions in $\mathcal{X}$, and approximate h by $\Phi_n'\beta$. Write $g_0(z) = E[Y \mid Z = z]$ and $\Pi_n(z) = E[\Phi_n(X) \mid Z = z]$. Let $\hat{g}_n$ and $\hat{\Pi}_n$ be nonparametric regression estimates of g_0 and Π_n respectively. For example, one can employ series regression: for a length-L_n column vector Ψ_n on S_Z, let $\hat{Q}_n = n^{-1}\sum_{i=1}^n \Psi_n(Z_i)\Psi_n(Z_i)'$. If $\hat{Q}_n$ is non-singular, the series estimates are given by

$$\hat{g}_n(z) := \Psi_n(z)'\hat{Q}_n^{-1} n^{-1} \sum_{i=1}^n \Psi_n(Z_i) Y_i, \tag{2.1}$$

$$\hat{\Pi}_n(z) := \left(n^{-1} \sum_{i=1}^n \Phi_n(X_i)\Psi_n(Z_i)' \right) \hat{Q}_n^{-1} \Psi_n(z). \tag{2.2}$$

Choose finite nonempty grids $\widetilde{W}_U \subseteq W_U$ and $\widetilde{S}_H \subseteq S_H$. The feasible constraints are

$$\mathbb{T}_U[\hat{g}_n - \hat{\Pi}_n'\beta](w) \le b(w), \qquad w \in \widetilde{W}_U, \tag{2.3}$$

$$\mathbb{T}_H[\Phi_n'](x)\beta \le C(x), \qquad x \in \widetilde{S}_H. \tag{2.4}$$

These are linear in β. The derivative example uses $|(\hat{g}_n - \hat{\Pi}_n'\beta)'(w)| \le b$ and therefore requires differentiable first-stage fits. The leading supremum example uses $|\hat{g}_n(z) - \hat{\Pi}_n(z)'\beta| \le b$, with $\widetilde{W}_U = \widetilde{S}_Z \subseteq S_Z$ the instrument grid. Here $\mathbb{T}_H[\Phi_n']$ is the matrix

obtained by applying $\mathbb{T}_H$ to each basis function. A grid generally approximates the continuum restrictions. Checking spline knots is sufficient only when the relevant piecewise polynomial attains its extrema there; this does not hold for arbitrary spline value constraints.

Let $\widehat{\mathbb{L}}_n$ be a linear functional estimating $\mathbb{L}$, with $\widehat{\mathbb{L}}_n = \mathbb{L}$ when it is known. The endpoint estimators are

$$\underline{\hat{\theta}}_{\mathbb{L}}(b) = \inf_{\beta \in \mathbb{R}^{K_n}} \widehat{\mathbb{L}}_n[\Phi_n'\beta] \quad \text{s.t. (2.3)–(2.4)}, \tag{2.5}$$

$$\hat{\bar{\theta}}_{\mathbb{L}}(b) = \sup_{\beta \in \mathbb{R}^{K_n}} \widehat{\mathbb{L}}_n[\Phi_n'\beta] \quad \text{s.t. (2.3)–(2.4)}. \tag{2.6}$$

For a known linear functional the objective is $\mathbb{L}[\Phi_n]'\beta$. For centered evaluation, use

$$\widehat{\mathbb{L}}_{x,n}[h] = h(x) - \frac{1}{n}\sum_i h(X_i), \tag{2.7}$$

so the sieve objective is

$$\left\{\Phi_n(x) - \frac{1}{n}\sum_i \Phi_n(X_i)\right\}'\beta.$$

After solving the programs, add $\bar{Y}_n := n^{-1}\sum_i Y_i$ to each endpoint and to their midpoint to obtain the normalized estimates. Thus the constraints remain (2.3)–(2.4). Each program has K_n variables and $d_U|\widetilde{W}_U| + d|\widetilde{S}_H|$ scalar inequalities. Theorem 3.2 includes the error from estimating $\mathbb{L}$.

For an average derivative $\mathbb{L}[h] = E[h'(X)]$ in the scalar case, use

$$\widehat{\mathbb{L}}_n[h] = n^{-1}\sum_i h'(X_i),$$

which again gives an objective linear in β.

For ordinary point evaluation, the objective is $\Phi_n(x)'\beta$; denote its endpoints by $\underline{\hat{\theta}}(x;b)$ and $\hat{\bar{\theta}}(x;b)$. The midpoint estimators are

$$\begin{aligned} \hat{\theta}^{\text{mid}}_{\mathbb{L}}(b) &= \tfrac{1}{2}\{\underline{\hat{\theta}}_{\mathbb{L}}(b) + \hat{\bar{\theta}}_{\mathbb{L}}(b)\}, \\ \hat{\theta}^{\text{mid}}(x;b) &= \tfrac{1}{2}\{\underline{\hat{\theta}}(x;b) + \hat{\bar{\theta}}(x;b)\}. \end{aligned} \tag{2.8}$$

An empty program is reported as infeasible at the chosen b; its midpoint is undefined. The theorem gives conditions ensuring feasibility with probability approaching one.

Appendix C studies the smallest relaxation b for which the maintained restrictions are feasible. It gives a sample analogue of this falsification frontier, its convergence rate, and a consistent misspecification test.

2.1 Calibrating b

The index b in (1.4) determines the size of the identified set. We suggest the researcher estimate the identified set for a range of choices for b corresponding to mild, moderate, and severe misspecification. To determine whether b corresponds to say, 'mild' misspecification, it is helpful to compare b to an estimable reference quantity. This approach is common in the empirical sensitivity literature. For example, Masten and Poirier (2018) compare the degree of selection on unobservables to selection on observables, which is estimable. See also Imbens (2003), Altonji et al. (2005), and Oster (2019).

For the leading supremum restriction (1.3), we suggest calibrating b against the residual variation in the outcome. Consider the following decomposition:

$$Y = E[h_0(X)|Z] + u_0(Z) + \eta$$

Under the additive direct-effect interpretation in Section 1, the first two terms capture the indirect and direct components associated with the instrument. More generally, u_0 aggregates departures from the conditional moment restriction. The residual η absorbs the effect of all other factors that influence Y. The variation in η is a useful reference quantity against which to calibrate the bound b on the magnitude of $u_0(Z)$. We measure the variation using a difference in quantiles.

Let $q_\eta(\alpha)$ denote the α quantile of η. We say b is small if it is equal to $q_\eta(0.5 + \tau) - q_\eta(0.5 - \tau)$ for a small τ. For example, $b = q_\eta(0.6) - q_\eta(0.4)$ corresponds to $\tau = 0.1$. In applications, the researcher can report identified sets for several values of τ, corresponding to mild, moderate, and severe departures from exact instrument validity. Note that $\eta = Y - E[Y|Z]$ is a reduced-form residual and thus we can estimate η and its quantiles in a first stage.

For a derivative bound, b is measured in outcome units per unit of the instrument. Calibration must therefore also specify the instrument scale; for example, multiplying

b by a specified change in Z gives an outcome-scale quantity for comparison with residual variation. Other choices of $\mathbb{T}_U$ similarly require a reference quantity in the units of their constraints.

3 Theoretical Analysis

Write $g_0(z) = E[Y \mid Z = z]$ and $\mathbb{A}[h](z) = E[h(X) \mid Z = z]$. We use versions in $\mathcal{Z}$ for $h \in \mathcal{X}$. The linear operator $\mathbb{A} : \mathcal{X} \to \mathcal{Z}$ satisfies $\|\mathbb{A}[h]\|_\infty \leq \|h\|_\infty$. Correct specification is the equation

$$g_0 = \mathbb{A}[h_0]. \tag{3.1}$$

All stochastic results concern an i.i.d. sample $\{(Y_i, X_i, Z_i)\}_{i=1}^n$. Function classes are separable for the suprema below, or probability statements are read in outer probability. The linear functionals $\mathbb{L}$ and $\widehat{\mathbb{L}}_n$ need not have bounded supremum-norm operator norms. The convergence results instead control functional values and sieve approximation on $\mathcal{H}$. This allows point and average derivatives when the shape restrictions provide the required control.

3.1 Midpoint estimation as a regularization scheme

If $\mathbb{A}$ is injective, correct specification identifies $h_0 = \mathbb{A}^{-1}[g_0]$. This inverse is discontinuous in an ill-posed NPIV problem. The midpoint instead targets the center of an identified interval under $g_0 - \mathbb{A}[h] \in \mathcal{U}_b$. Positive slack in the inequalities makes that target locally stable.

Let $\widetilde{g}$ and $\widetilde{\mathbb{A}}$ replace g_0 and $\mathbb{A}$, and let a tilde on an endpoint denote its value with these replacements, at the same b, $\mathcal{H}$, and $\mathbb{L}$. Set

$$\eta = \sup_{h \in \mathcal{H}} \|\mathbb{T}_U[(\widetilde{g} - \widetilde{\mathbb{A}}[h]) - (g_0 - \mathbb{A}[h])]\|_\infty.$$

Proposition 3.1 (Stability of endpoints and midpoints)**.** *Suppose $\mathcal{H}$ is convex and $h^* \in \mathcal{H}$ satisfies*

$$\mathbb{T}_U[g_0 - \mathbb{A}[h^*]](w) \leq \epsilon(w) < b(w), \qquad w \in W_U,$$

for a vector function ϵ, where the componentwise reciprocal $(b-\epsilon)^{-1}$ is bounded and $\eta\|(b-\epsilon)^{-1}\|_\infty < 1$. The supremum norm is over w and all components, so boundedness requires a positive uniform gap between ϵ and b. If the population endpoints for $\mathbb{L}$ are finite, the perturbed endpoints are finite and

$$|\widetilde{\bar{\theta}}_{\mathbb{L}}(b) - \bar{\theta}_{\mathbb{L}}(b)| \le \eta\|(b-\epsilon)^{-1}\|_\infty\{\bar{\theta}_{\mathbb{L}}(b) - \mathbb{L}[h^*]\},$$
$$|\widetilde{\underline{\theta}}_{\mathbb{L}}(b) - \underline{\theta}_{\mathbb{L}}(b)| \le \eta\|(b-\epsilon)^{-1}\|_\infty\{\mathbb{L}[h^*] - \underline{\theta}_{\mathbb{L}}(b)\}.$$

Consequently,

$$|\widetilde{\theta}^{\mathrm{mid}}_{\mathbb{L}}(b) - \theta^{\mathrm{mid}}_{\mathbb{L}}(b)| \le \frac{\eta}{2}\|(b-\epsilon)^{-1}\|_\infty\{\bar{\theta}_{\mathbb{L}}(b) - \underline{\theta}_{\mathbb{L}}(b)\}.$$

These statements hold uniformly over point evaluations when the corresponding widths are uniformly bounded.

For $\|u\|_\infty \le b$, an anchor with $\|g_0 - \mathbb{A}[h^*]\|_\infty \le \varepsilon_0 < b$ allows $\epsilon(w) = (\varepsilon_0, \varepsilon_0)'$, giving $\|(b-\epsilon)^{-1}\|_\infty = (b-\varepsilon_0)^{-1}$. Under correct specification, $h^* = h_0$ and $\epsilon = 0$ give $1/b$. For the derivative restriction, taking $\epsilon = \mathbb{T}_U[g_0 - \mathbb{A}[h^*]]$ gives the multiplier $\{b - \|(g_0 - \mathbb{A}[h^*])'\|_\infty\}^{-1}$ when the gap is positive. The argument is the same convex-mixture repair in each case. Estimation of the target functional is handled separately below.

3.2 Consistency and convergence rates

Fix $\bar{b} > 0$. The same first-stage estimates, sieve, and grids are used for every $b \in [0, \bar{b}]$. Let $\mathcal{H}_n = \mathcal{H} \cap \mathrm{span}(\Phi_n)$ and $\Pi_n(z) = E[\Phi_n(X) \mid Z = z]$. Define

$$D_{H,n} = \sup_{x \in S_H} \min_{x' \in \widetilde{S}_H} \|x - x'\|_2, \qquad D_{U,n} = \sup_{w \in W_U} \min_{w' \in \widetilde{W}_U} \|w - w'\|_2,$$

and, for a linearly independent basis,

$$\iota_n = 1 + \sup_{\beta \neq 0} \frac{\|\beta\|_2}{\|\Phi_n'\beta\|_\infty} < \infty.$$

Assumption 3.1 (Transformed first-stage error)**.** For a deterministic $a_n \downarrow 0$,

$$\eta_n := \sup_{\beta:\, \Phi_n'\beta \in \mathcal{H}} \|\mathbb{T}_U[(\hat{g}_n - \hat{\Pi}_n'\beta) - (g_0 - \Pi_n'\beta)]\|_\infty = O_p(a_n).$$

Assumption 3.2 (Sieve approximation)**.** There is a sequence $\kappa_n \to 0$ and approximants $q_n[h] \in \mathcal{H}_n$ such that, for every fixed $R < \infty$,

$$\sup_{h \in \mathcal{H}:\, \|h\|_\infty \le R} \left\{ \|q_n[h] - h\|_\infty + |\mathbb{L}[q_n[h]] - \mathbb{L}[h]| \right\} = O(\kappa_n).$$

Additionally, $0 \in \mathcal{H}$ and each component of $C(x)$ is bounded below away from zero uniformly over $x \in S_H$.[4]

Assumption 3.3 (Grid regularity)**.** Both $C(\cdot)$ and each row of $\mathbb{T}_H[\Phi_n']$ are Lipschitz with constants at most ξ_n. Uniformly over $b \in [0, \bar{b}]$, $b(\cdot)$ has Lipschitz constant at most G_n. With probability approaching one, the same bound holds for $\mathbb{T}_U[\hat{g}_n]$ and every row of $\mathbb{T}_U[\hat{\Pi}_n']$. Both mesh widths tend to zero and

$$d_n := \iota_n \xi_n D_{H,n} + \iota_n G_n D_{U,n} \longrightarrow 0.$$

The grid argument uses only these mesh widths and Lipschitz bounds; neither S_H nor W_U needs to be a rectangle.

Assumption 3.4 (Functional values and estimation)**.** For every fixed $R < \infty$,

$$\begin{aligned} \sup_{h \in \mathcal{H}:\, \|h\|_\infty \le R} |\mathbb{L}[h]| &< \infty, \\ \sup_{h \in \mathcal{H}_n:\, \|h\|_\infty \le R} |\widehat{\mathbb{L}}_n[h] - \mathbb{L}[h]| &= O_p(a_{\mathbb{L},n}), \end{aligned} \tag{3.2}$$

where $\widehat{\mathbb{L}}_n$ is linear and $a_{\mathbb{L},n} \to 0$ is a deterministic nonnegative sequence.

When $\mathbb{L}$ is known, take $\widehat{\mathbb{L}}_n = \mathbb{L}$: the estimation-error condition holds trivially with $a_{\mathbb{L},n} = 0$. For uniform results over a family of functionals, impose the bounds in Assumptions 3.2 and 3.4 uniformly, using common approximants $q_n[h]$ and rates.

For the transformed restrictions, also suppose

$$\|\mathbb{T}_U \mathbb{A}[h]\|_\infty \le C_A \|h\|_\infty, \qquad h \in \mathcal{X}, \tag{3.3}$$

[4] The conditions on 0 and C can be relaxed to the existence of $h^s \in \mathcal{H}$ and $q_n^s \in \mathcal{H}_n$ such that $\|q_n^s - h^s\|_\infty \le \kappa_n$ and $\mathbb{T}_H[q_n^s](x) \le C(x) - \sigma \mathbf{1}$ for all $x \in S_H$, for some fixed $\sigma > 0$. The proof of Theorem 3.2 uses this more general condition.

with $C_A < \infty$. This bounds the composition with the conditional expectation operator, and holds with $C_A = 1$ in the leading case. For derivative constraints it follows, for example, from suitable smoothness of the conditional density. Set $r_n = a_n + \kappa_n + d_n$.

Theorem 3.2 (A rate uniform in the relaxation index)**.** *Suppose Assumptions 3.1–3.4 and (3.3) hold. Fix $h^* \in \mathcal{H}$ and a vector function ϵ, both independent of b and n. Consider any deterministic $\underline{b}_n \in (0, \bar{b}]$ such that, for every $b \in [\underline{b}_n, \bar{b}]$,*

$$\mathbb{T}_U[g_0 - \mathbb{A}[h^*]](w) \leq \epsilon(w) < b(w), \qquad w \in W_U. \tag{3.4}$$

Suppose $(b - \epsilon)^{-1}$ is bounded, with reciprocals taken componentwise as in Proposition 3.1, and set

$$M := \sup_{h \in \Theta_{\bar{b}}} \|h\|_\infty < \infty, \qquad s(b) := \min\{1, \|(b - \epsilon)^{-1}\|_\infty^{-1}\}. \tag{3.5}$$

The bounds $b(\cdot)$ are nondecreasing in the scalar index and ϵ is fixed, so $s(b)$ is nondecreasing. If $r_n = o(s(\underline{b}_n))$, the sample programs are simultaneously nonempty and bounded for $b \in [\underline{b}_n, \bar{b}]$ with probability approaching one. Moreover,

$$\sup_{b \in [\underline{b}_n, \bar{b}]} \frac{\max\left\{|\hat{\underline{\theta}}_{\mathbb{L}}(b) - \underline{\theta}_{\mathbb{L}}(b)|, |\hat{\bar{\theta}}_{\mathbb{L}}(b) - \bar{\theta}_{\mathbb{L}}(b)|\right\}}{r_n / s(b) + a_{\mathbb{L},n}} = O_p(1). \tag{3.6}$$

The same rate holds for midpoints, and uniformly over any family satisfying the uniform versions of Assumptions 3.2 and 3.4.

Objective estimation contributes an additive error. For normalized estimates, adding $\bar{Y}_n$ contributes the further error $|\bar{Y}_n - E[Y]| = O_p(n^{-1/2})$ when $E[Y^2] < \infty$. The same observations may be used for the objective and first stages.

The uniform bound also applies to a data-dependent b in the stated interval with probability approaching one. Uniformity is restricted to bounds $b(\cdot)$ strictly above the fixed $\epsilon(\cdot)$, with the gap controlled by $r_n = o(s(\underline{b}_n))$. For the leading restriction and a fixed anchor with residual norm at most ε_0, use $\epsilon(w) = (\varepsilon_0, \varepsilon_0)'$ for $b > \varepsilon_0$, giving $s(b) = \min\{1, b - \varepsilon_0\}$. Under correct specification, known-objective errors are $O_p(r_n/b)$ near zero. A derivative bound has the same slack order when the anchor residual is constant, but its first-stage rate must control derivatives.

Boundedness of $\Theta_{\bar{b}}$ holds if $\mathcal{H}$ is bounded, but can also hold for an unbounded $\mathcal{H}$. The proof derives boundedness of all grid-feasible functions from this condition and the anchors. Dense grids reduce d_n, while increasing the sieve dimension may reduce κ_n and increase ι_n or ξ_n.

Corollary 3.3 (Midpoint convergence as $b \downarrow 0$)**.** *Suppose the theorem's conditions hold for the same h^* and ϵ, with $s(b) > 0$ for every $b \in (0, \bar{b}]$. Assume $\overline{\mathcal{H}}$ is compact in supremum norm and for every w, $b(w)$ is right-continuous in the scalar index at zero. Let $h_0 \in \Theta_0$. Suppose $\mathbb{L}$ has a continuous extension from $\mathcal{H}$ to $\overline{\mathcal{H}}$ in the supremum norm, denoted by the same symbol, and is identified at zero tolerance: every $h \in \overline{\mathcal{H}}$ satisfying the moment inequalities at index zero has $\mathbb{L}[h] = \mathbb{L}[h_0]$. Then*

$$\omega_{\mathbb{L}}(b) := \sup_{h \in \Theta_b} |\mathbb{L}[h] - \mathbb{L}[h_0]| \longrightarrow 0, \qquad |\theta^{\mathrm{mid}}_{\mathbb{L}}(b) - \mathbb{L}[h_0]| \leq \omega_{\mathbb{L}}(b).$$

For $b_n \downarrow 0$ with $r_n = o(s(b_n))$,

$$|\hat{\theta}^{\mathrm{mid}}_{\mathbb{L}}(b_n) - \mathbb{L}[h_0]| \leq \omega_{\mathbb{L}}(b_n) + O_p(r_n/s(b_n) + a_{\mathbb{L},n}) = o_p(1). \tag{3.7}$$

These conclusions hold uniformly over families equicontinuous on $\overline{\mathcal{H}}$ and satisfying the stated identification property and the uniform versions of the theorem's assumptions. If $\omega_{\mathbb{L}}(b) \leq Cb^{\alpha}$ and $s(b) \geq cb^{\nu}$ near zero, $\alpha, \nu > 0$, the choice $b_n \asymp r_n^{1/(\alpha+\nu)}$ gives rate $O_p(r_n^{\alpha/(\alpha+\nu)} + a_{\mathbb{L},n})$.

Closures here are taken among bounded continuous functions in the supremum norm. The operators $\mathbb{A}$ and $\mathbb{T}_U\mathbb{A}$ are used with their continuous extensions to these closures, justified by their norm bounds. For the functional, the corollary requires continuity only on $\overline{\mathcal{H}}$. For the supremum restriction, $\mathcal{U}_0 = \{0\}$ and completeness gives the identification condition for point evaluation, recovering convergence to h_0 uniformly in x.

For the derivative-only restriction, $\mathcal{U}_0$ contains constants. If constants belong to $\mathcal{X}$ and $\mathbb{A}$ is injective on its supremum-norm closure, any two zero-tolerance feasible functions differ by a constant. Thus the centered evaluation $\mathbb{L}_x$ is identified even when their levels are not. The corollary gives uniform convergence of its midpoint estimator to $h_0(x) - E[h_0(X)]$ when Assumption 3.4 holds uniformly in x. Adding $\bar{Y}_n$ gives uniform convergence to $h_0(x) + E[Y] - E[h_0(X)]$, the normalized structural

function, provided $\bar{Y}_n \to_p E[Y]$. For this example,

$$\|u - E[u(Z)]\|_\infty \leq b, \qquad u \in \mathcal{U}_b,$$

on $[0,1]$, because $|u(z) - u(z')| \leq b|z - z'|$. Hence the centered residual vanishes as $b \downarrow 0$, although the unrestricted constant part of u need not vanish.

Appendix D gives primitive conditions for the first stage and sieve approximation, including the requirements for point and average first derivatives under range and curvature restrictions.

4 Application

The reported empirical results use the leading special case $\mathcal{U}_b = \{u : \|u\|_\infty \leq b\}$ and ordinary point evaluation $\mathbb{L}[h] = h(x)$. To demonstrate our methods, we apply them to the setting studied in Section 5.1 of Horowitz (2011). Horowitz (2011) estimates a shape-invariant Engel curve for food using data from the British Family Expenditure Survey.[5] Horowitz's application is in turn based on Blundell et al. (2007) who also carry out NPIV estimation of shape-invariant Engel curves and use the same data.

Endogeneity. In observational settings, the share of household wealth allocated to nondurables is decided by the household. Therefore, the budget for nondurables depends on household preferences. The same preferences also influence the allocation of the nondurable budget across categories such as food.

To tackle this problem Blundell et al. (2007) and Horowitz (2011) use household income as an instrument for nondurables expenditure. There are many reasons to think that household income and preferences are correlated. Both tastes and income may depend on household size and socio-economic status, and those with expensive tastes may seek high-paying jobs. However, some shocks to household income may result from exogenous external factors like shocks to production costs in an employed householder's firm. If one controls for a sufficiently rich set of household covariates this may absorb the endogenous variation in income leaving only the exogenous variation.

[5] We made use of the data file that accompanies Horowitz (2011) and adapted the accompanying code in order to evaluate Horowitz's estimator and B-spline bases for our own methods.

If the data do not contain rich enough household observables, then some endogeneity likely remains and the income instrument is unlikely to be fully valid.

Blundell et al. (2007) and Horowitz (2011) can only control for a limited set of demographic variables.[6] Therefore, in this setting instrumental validity may not hold exactly and it is of interest to assess what empirical findings are robust to some failure of instrumental validity.

Data and variables. The sample consists of $n = 1655$ households drawn from the British Family Expenditure Survey. Following Horowitz (2011), we restrict attention to married couples with an employed head-of-household aged 25–55. Let Y_i be the share of non-durable expenditure spent on food, let X_i be the logarithm of total non-durable expenditure, and let Z_i be the logarithm of household income. Our object of interest is the function $h_0 : S_X \to [0, 1]$, where S_X denotes the support of X.

Shape restrictions and function class. We impose bounded curvature and range restrictions by taking

$$\mathcal{H}(c) = \Big\{ h \in C^2(S_X) \ : \ 0 \le h \le 1, \ \sup_{x \in S_X} |h''(x)| \le c \Big\}.$$

We report results for $c = 1.0$, $c = 1.25$ and $c = 1.5$ representing tighter, moderate, and looser curvature constraints. For comparison, the magnitude of the second derivative of Horowitz's estimated structural function never exceeds 0.5.

In the computation, X is normalized to the unit interval before constructing the spline basis. The curvature bounds reported are stated with respect to the original log-expenditure scale.[7]

Calibrating departures from instrument validity. Following the calibration approach in Section 2.1, we index the magnitude of misspecification by a scalar b. In particular, for a given $\tau \in (0, 1/2)$, we set

$$\hat{b}(\tau) = \hat{q}_\eta(0.5 + \tau) - \hat{q}_\eta(0.5 - \tau),$$

[6] Both papers control for demographics by analyzing a homogeneous sub-sample. Blundell et al. (2007) incorporate additional dummy variable controls.

[7] Let $x^{\text{norm}} = \frac{x^{\text{orig}} - \min(x^{\text{orig}})}{\Delta_x}$ be the normalized X with $\Delta_x = \max(x^{\text{orig}}) - \min(x^{\text{orig}})$. A curvature bound c on the original log-expenditure scale is implemented as the bound $c\Delta_x^2$.

where $\hat{q}_\eta(\cdot)$ denotes the empirical quantile function of residuals $\hat{\eta}_i = Y_i - \hat{g}_n(Z_i)$. Here $\hat{g}_n$ is a nonparametric estimate of the reduced-form function $g_0(z) = \mathbb{E}[Y \mid Z = z]$ (implemented via a spline series regression in Z). We consider $\tau \in \{0.05, 0.1, 0.15\}$, which we interpret as mild, moderate, and severe deviations from exact instrument validity.[8]

Sieve approximation and cross-validation tuning. Horowitz (2011) uses cubic B-spline bases with evenly spaced knots for the structural sieve Φ_n.[9] We also use spline bases, but allow the first-stage and second-stage basis dimensions to differ. For the reported finite-sample calculation, we set the dimension of Φ_n to $K_n = 10$. For the first stage, we choose the dimension L_n of the Z-basis jointly with a ridge (Tikhonov) regularization parameter by cross-validation. Specifically, we consider a grid of candidate spline dimensions $L \in \mathcal{L}$ and ridge penalties $\lambda \in \Lambda$. We then use the one-standard-error cross-validation rule described in Appendix B to select (L, λ).

Grids and implementation choices. Taking $S_H = S_X$, we evaluate the pointwise identified intervals for $h_0(x)$ on a grid $\widetilde{S}_H$ of 80 equally spaced points between the minimum and maximum observed values of X. We construct $\widetilde{S}_Z$ as 80 equally spaced points between the 0.005 and 0.995 empirical quantiles of Z. These are the constraints used for the reported computation. Trimming the instrument grid does not by itself approximate a moment restriction on the full support; applying the full-support theory requires control of the omitted tails or an explicitly restricted instrument domain.

Sampling uncertainty. The reported calculation uses subsamples of size $m = \lfloor n^{0.75} \rfloor$, drawn without replacement, with $B = 1000$ replications. It holds the calibrated $\hat{b}(\tau)$ fixed across subsamples. We describe the calculation explicitly, while distinguishing it from an established coverage result.

A standard justification for subsampling uses a suitable normalizing sequence $\rho_n \to \infty$ and a limiting distribution for the normalized estimation error, together with $m \to \infty$, $m/n \to 0$, $\rho_m/\rho_n \to 0$, negligible Monte Carlo error, and the corresponding

[8] In the implementation, the implied values of $\hat{b}(\tau)$ are computed directly from the sample residuals $\hat{\eta}_i$ using the definition above.

[9] See de Boor (2014) for a practical introduction to B-splines.

bias and quantile-continuity conditions (Politis and Romano, 1994). A convergence-rate bound alone does not supply these conditions. In a growing nonparametric sieve, ρ_n is generally not $\sqrt{n}$. Neither such a limit law nor negligibility of sieve, grid, ridge, and calibration errors at the inference scale is established here. Holding $\hat{b}(\tau)$ fixed also does not automatically account for its sampling variation.

The formulas below use a generic ρ_n; the calculations in Figure 4.1 use $\rho_n = \sqrt{n}$. Its shaded regions should therefore be interpreted as illustrative sampling-uncertainty summaries, with nominal coverage requiring separate justification for that implementation.

The pointwise identified intervals have lower and upper envelope functions

$$\underline{\theta}(x) = \inf_{h \in \Theta_b} h(x), \qquad \bar{\theta}(x) = \sup_{h \in \Theta_b} h(x).$$

We first use subsampling to approximate the sampling uncertainty in the estimated envelopes $\hat{\underline{\theta}}(x)$ and $\hat{\bar{\theta}}(x)$. For subsample $s \in \{1, \ldots, B\}$, define

$$T_{\ell,s} = \sup_{x \in \widetilde{S}_H} \rho_m \left| \hat{\underline{\theta}}^{(s)}(x) - \hat{\underline{\theta}}(x) \right|,$$

and

$$T_{u,s} = \sup_{x \in \widetilde{S}_H} \rho_m \left| \hat{\bar{\theta}}^{(s)}(x) - \hat{\bar{\theta}}(x) \right|.$$

Let $q_{\ell,1-\alpha/2}$ and $q_{u,1-\alpha/2}$ denote the empirical $(1-\alpha/2)$-quantiles of $\{T_{\ell,s}\}_{s=1}^B$ and $\{T_{u,s}\}_{s=1}^B$, respectively.

The corresponding Bonferroni-adjusted candidate bands for the two envelope functions are:

$$\widehat{\mathcal{C}}_\ell(x) = \left[\hat{\underline{\theta}}(x) - \frac{q_{\ell,1-\alpha/2}}{\rho_n},\ \hat{\underline{\theta}}(x) + \frac{q_{\ell,1-\alpha/2}}{\rho_n} \right],$$

and

$$\widehat{\mathcal{C}}_u(x) = \left[\hat{\bar{\theta}}(x) - \frac{q_{u,1-\alpha/2}}{\rho_n},\ \hat{\bar{\theta}}(x) + \frac{q_{u,1-\alpha/2}}{\rho_n} \right], \qquad x \in \widetilde{S}_H.$$

Under the distributional and bias conditions just described, these would provide simultaneous coverage of the two envelopes over the evaluation grid.

Because the maintained assumptions imply

$$\underline{\theta}(x) \le h_0(x) \le \bar{\theta}(x), \qquad x \in S_X,$$

simultaneously valid envelope bands would imply an outer band for the structural function:

$$\widehat{\mathcal{C}}_h(x) = \left[\max\left\{0, \underline{\hat{\theta}}(x) - \frac{q_{\ell,1-\alpha/2}}{\rho_n}\right\},\ \min\left\{1, \hat{\bar{\theta}}(x) + \frac{q_{u,1-\alpha/2}}{\rho_n}\right\}\right], \qquad x \in \widetilde{S}_H.$$

The truncation by 0 and 1 uses the maintained range restriction $h_0(x) \in [0,1]$. If both envelope bands have their stated asymptotic coverage, Bonferroni gives coverage at least $1-\alpha$ for this outer band over the evaluation grid. The present rate results do not establish that premise for the reported calculation.

We also report the midpoint estimator

$$\hat{\theta}^{\text{mid}}(x) = \frac{\underline{\hat{\theta}}(x) + \hat{\bar{\theta}}(x)}{2},$$

which provides the pointwise minimax summary described in Section 1.3. This does not assert minimaxity under integrated squared loss over the joint identified set.

Results. Figure 4.1 reports the estimated identified sets for $h_0(x)$ across combinations of the misspecification calibration parameter τ and the curvature bound c. Columns correspond to increasing values of $\tau \in \{0.05, 0.10, 0.15\}$, while rows correspond to increasing values of $c \in \{1.0, 1.25, 1.5\}$. In each panel, the red and blue solid curves are the estimated lower and upper envelopes, $\underline{\hat{\theta}}(x)$ and $\hat{\bar{\theta}}(x)$, respectively. The red and blue shaded regions are Bonferroni-adjusted subsampling uncertainty bands for these two envelope functions. The black dash-dotted curves give the implied outer uncertainty band for the structural function $h_0(x)$. The green solid curve is the midpoint estimator $\hat{\theta}^{\text{mid}}(x)$, and the gray dashed curve reproduces the NPIV point estimate reported by Horowitz (2011).

The figure illustrates the robustness–smoothness trade-off emphasized in the theory. Holding c fixed, moving from left to right increases τ, and therefore increases the calibrated bound $b(\tau)$ on deviations from the exact NPIV moment condition. This weakens the maintained instrument-validity restriction and enlarges the estimated identified set. Holding τ fixed, moving from top to bottom increases c, thereby weakening the curvature restriction and expanding the class of admissible structural functions. Thus the width of the estimated set reflects both the allowed degree of instrument invalidity and the strength of the maintained smoothness restriction.

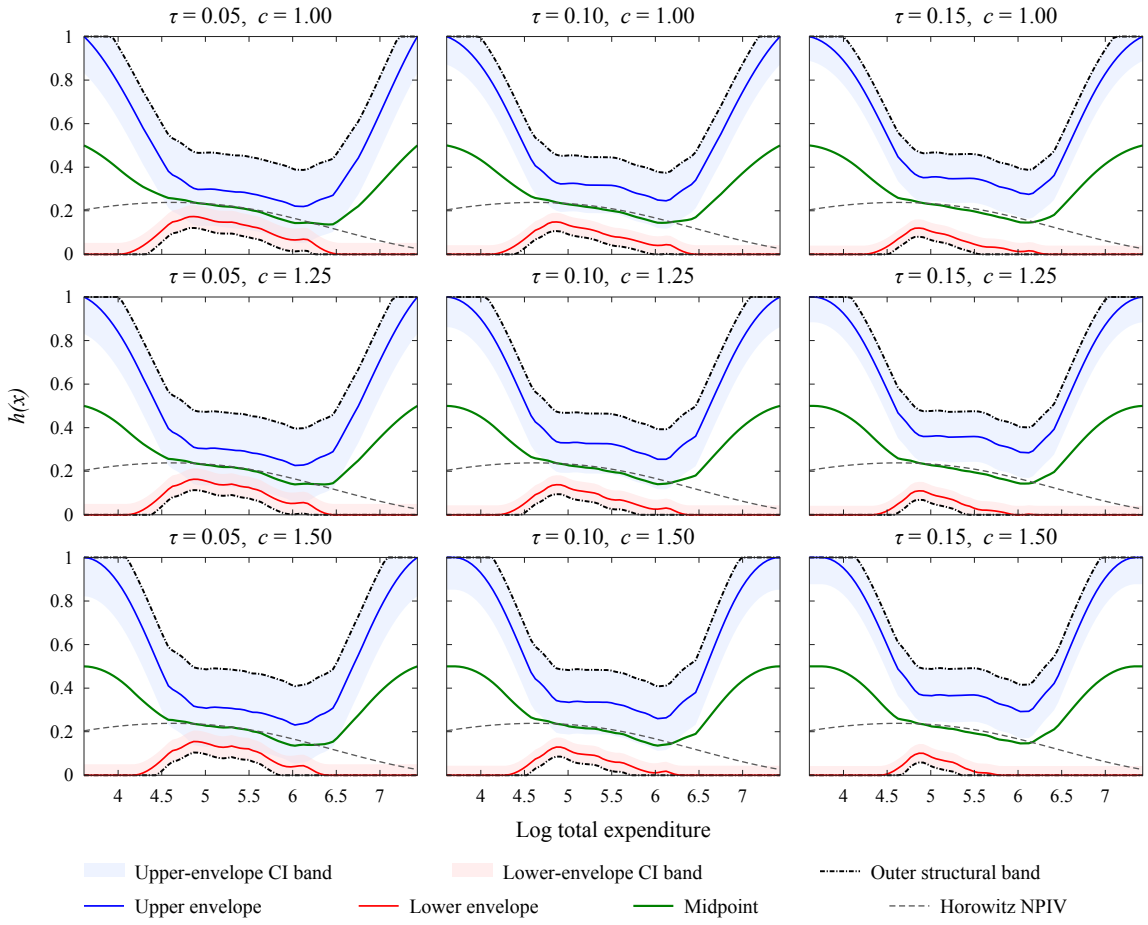


Figure 4.1: Set-estimated Engel curves across misspecification levels and curvature restrictions.

Notes: Columns correspond to $\tau \in \{0.05, 0.10, 0.15\}$, and rows correspond to $c \in \{1.0, 1.25, 1.5\}$. The red and blue solid curves are the estimated lower and upper envelopes of the identified set for $h_0(x)$. The red and blue shaded regions are Bonferroni-adjusted subsampling uncertainty bands for the lower and upper envelope functions. The black dash-dotted curves are the implied outer uncertainty band for $h_0(x)$, obtained from the lower edge of the lower-envelope band and the upper edge of the upper-envelope band. The green solid curve is the midpoint estimator $\hat{\theta}^{\text{mid}}(x)$. The gray dashed curve is the NPIV estimate from Horowitz (2011). The first-stage reduced-form regressions use the cross-validated tuning parameters $L_n = 4$ and $\lambda = 2.85 \times 10^{-9}$. Subsampling uses $m = \lfloor n^{0.75} \rfloor$ and $B = 1000$, with the reported root-n normalization. Nominal coverage of these illustrative bands is not established by the rate theory.

The midpoint and Horowitz's estimate decline over much of the interior expenditure range, while the pointwise bounds are wide near its extremes. The figure alone does not establish whether the low-expenditure upturn emphasized by Horowitz (2011) is robust to misspecification. That question requires optimizing the relevant contrasts over the common feasible set, or checking whether that set contains a nonincreasing function. A curve lying between the pointwise envelopes need not itself be feasible.

5 Simulations

We first examine the performance of the robust point estimator in correctly specified models. We simulate data from five univariate NPIV models used in Kankanala (2025), following Newey and Powell (2003) (NP), Santos (2012) (S), Chernozhukov et al. (2023) (CNS), Chetverikov and Wilhelm (2017) (CW), and Chen et al. (2025) (CCK). The structural functions are, respectively, a signed logarithm, a sine function, a decreasing normal distribution function, a piecewise quadratic, and a product of a sine function and a logarithm. These baseline models satisfy $E[Y - h_0(X) \mid Z] = 0$. The exercise therefore assesses the estimator's regularization properties when the instruments are valid. Appendix A specifies the models and implementation.

For each model, we generate $R = 1000$ samples at each of $n = 1000$ and $n = 10{,}000$. We approximate h by a ten-dimensional cubic spline and estimate the conditional expectations by unpenalized, ten-dimensional thin-plate spline regressions, also used for QB below. We impose the second derivative bound $|h''| \leq c$, with no range restriction. We set $c = ac_0$ with $a = 1.5$, where c_0 is the supremum of the absolute second derivative of h_0 for NP, S, CNS, and CW, taken over points where it exists. The second derivative is undefined at $x = 1$ in NP and at $x = 1/2$ in CW. For CCK, the maximum of the second derivative is infinite, and so we instead set $c_0 = 8$. Thus the values of c_0 are 1, $2\pi^2$, $f_N(1/2)$, 4, and 8 for NP, S, CNS, CW, and CCK, respectively, where f_N is the standard normal density. Appendix A provides results for $c = ac_0$ for various values of a. These restrictions and the conditional moment inequalities are imposed at 101 quantiles of X and Z, respectively. We initialize b at the difference between the 55th and 45th percentiles of $Y - g_0(Z)$, where $g_0(Z) = E[Y \mid Z]$, and increase it if needed for successful optimization. Here

we use the population quantiles; an empirical implementation could use quantiles of estimated regression residuals.

Each replication is evaluated at $M = 1000$ new draws from the untrimmed distribution of X. We report

$$\widehat{\mathcal{R}} = \left\{ \frac{1}{RM} \sum_{r=1}^{R} \sum_{j=1}^{M} \left(\widehat{h}_r^{\text{mid}}(X'_{rj}) - h_0(X'_{rj}) \right)^2 \right\}^{1/2}. \tag{5.1}$$

This averages squared errors before taking the square root, as in Kankanala (2025). We also report mean absolute error, calculated by averaging $|\widehat{h}_r(X'_{rj}) - h_0(X'_{rj})|$ over the same replications and test observations. Table 1 compares the midpoint at $a = 1.5$ with series two-stage least squares (series 2SLS), ordinary linear two-stage least squares (linear 2SLS), and the quasi-Bayes posterior mean (QB) of Kankanala (2025). All four estimators use the same training samples and test covariates. Series 2SLS uses a thin-plate spline first stage and a natural-spline structural function, both with dimension $J = 3$, the choice with the smallest published risk in all five models. Linear 2SLS regresses Y on an intercept and X, using an intercept and Z as instruments. For QB we use a Whittle–Matérn Gaussian process prior and a ten-dimensional thin-plate first stage at both sample sizes. This dimension attains the smallest published QB risk among $K \in \{5, 7, 10\}$ in all five models, with a tie in CNS. Appendix A describes our QB implementation and compares its $n = 1000$ results with the published values.

At $n = 1000$, series 2SLS has the lowest errors under both loss criteria in NP, QB in S and CCK, and linear 2SLS in CNS and CW. At $n = 10{,}000$, QB has the lowest errors in NP, S, and CCK, and linear 2SLS in CNS. In CW at the larger sample size, the midpoint has the lowest root mean squared error, while series 2SLS has the lowest mean absolute error. Increasing the sample size reduces both midpoint errors in NP, S, CNS, and CW. In CCK, its root mean squared error is nearly unchanged and its mean absolute error rises slightly.

Our $J = 3$ series 2SLS risks are close to the published values in NP, S, CNS, and CCK. In CW, our risk is 3.145, compared with the published 1.623. Additional simulations show substantial variation across batches because a few samples yield very large estimation errors. Appendix A discusses this discrepancy, compares both series 2SLS and QB with the published results, and examines sensitivity of the midpoint to the second derivative bound.

Table 1: Root mean squared and mean absolute errors

Model	Midpoint	Series 2SLS	Linear 2SLS	QB
$n = 1000$				
NP	0.155 (0.107)	0.128 (0.089)	0.172 (0.141)	0.135 (0.097)
S	0.267 (0.192)	0.293 (0.208)	0.257 (0.186)	0.177 (0.131)
CNS	0.101 (0.074)	0.195 (0.139)	0.074 (0.054)	0.145 (0.104)
CW	0.110 (0.086)	3.145 (0.405)	0.098 (0.078)	0.122 (0.094)
CCK	0.311 (0.257)	0.349 (0.297)	0.336 (0.281)	0.261 (0.213)
$n = 10{,}000$				
NP	0.080 (0.063)	0.102 (0.071)	0.164 (0.137)	0.065 (0.048)
S	0.230 (0.178)	0.252 (0.183)	0.249 (0.181)	0.104 (0.075)
CNS	0.041 (0.029)	0.057 (0.042)	0.023 (0.017)	0.081 (0.058)
CW	0.081 (0.068)	0.085 (0.063)	0.082 (0.069)	0.094 (0.076)
CCK	0.310 (0.260)	0.283 (0.249)	0.321 (0.273)	0.169 (0.136)

Notes: Entries are root mean squared errors, with mean absolute errors in parentheses. All four estimators use the same $R = 1000$ training samples and $M = 1000$ test covariates per replication. The midpoint uses $a = 1.5$, no range restriction, and the population 55th–45th percentile gap of $Y - g_0(Z)$ to initialize b. Series 2SLS uses $J = 3$ in both stages; linear 2SLS regresses Y on $(1, X)$ using $(1, Z)$ as instruments. QB is our implementation of the posterior mean in Kankanala (2025), using the same ten-dimensional first-stage regression as the midpoint. Appendix A gives implementation details and compares its risk with the published results. Monte Carlo standard errors for the midpoint, linear 2SLS, and QB are all below 0.005 for both loss criteria.

We also examine misspecified versions of the same models. We replace Y by $Y + \frac{1}{2} b_{\text{initial}} \cos(Z)$, where

$$b_{\text{initial}} = q_\eta(0.60) - q_\eta(0.40), \qquad \eta = Y - E[Y \mid Z].$$

Thus $u_0(z) = \frac{1}{2} b_{\text{initial}} \cos(z)$. Adding a function of Z leaves the distribution of η unchanged. The midpoint estimator starts at the full b_{initial} and retains the same rule for increasing it after unsuccessful optimization. Table 2 reports results using the same simulation draws, with all four estimators refitted to the modified outcomes. Errors are measured relative to the original h_0; all other settings are unchanged.

In Table 2, the midpoint has lower errors than QB under both loss criteria in NP, CNS, and CW at both sample sizes. In S, it has a lower mean absolute error but a higher root mean squared error. At $n = 10{,}000$, the midpoint has the lowest errors among all four estimators in CNS and CW, while linear 2SLS performs best in NP. In CCK, QB performs best and the midpoint has the highest errors under both criteria at both sample sizes.

Table 2: Root mean squared and mean absolute errors under misspecification

Model	Midpoint	Series 2SLS	Linear 2SLS	QB
$n = 1000$				
NP	0.368 (0.314)	0.391 (0.337)	0.268 (0.234)	0.403 (0.346)
S	0.503 (0.445)	0.539 (0.476)	0.514 (0.461)	0.483 (0.453)
CNS	0.232 (0.204)	0.296 (0.245)	0.227 (0.207)	0.270 (0.226)
CW	0.186 (0.167)	2.465 (0.450)	0.181 (0.166)	0.193 (0.172)
CCK	0.534 (0.424)	0.510 (0.403)	0.500 (0.388)	0.452 (0.351)
$n = 10{,}000$				
NP	0.348 (0.307)	0.380 (0.331)	0.263 (0.227)	0.401 (0.354)
S	0.486 (0.437)	0.516 (0.470)	0.509 (0.459)	0.464 (0.446)
CNS	0.205 (0.194)	0.225 (0.207)	0.215 (0.203)	0.242 (0.210)
CW	0.168 (0.157)	0.191 (0.168)	0.171 (0.162)	0.178 (0.162)
CCK	0.550 (0.415)	0.459 (0.348)	0.486 (0.367)	0.396 (0.313)

Notes: Entries are root mean squared errors, with mean absolute errors in parentheses, relative to the original structural function h_0. Each model and sample size uses $R = 1000$ replications and $M = 1000$ independent test covariates per replication. The midpoint uses $a = 1.5$, no range restriction, and the population 60th–40th percentile gap of $Y - E[Y \mid Z]$ to initialize b. The added conditional moment violation is half this initial gap times $\cos(Z)$. For CW at $n = 1000$, one replication accounts for approximately 73% of the aggregate squared error of series 2SLS, explaining its large root mean squared error.

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A Simulation models and implementation

A.1 Models

The correctly specified models have $Y = h_0(X)+\varepsilon$. Write F_N and f_N for the standard normal distribution and density, and $t_+ = \max\{t, 0\}$. The following specifications reproduce the univariate NPIV designs in Appendix A of Kankanala (2025).

NP. Let (ε, V, Z) be jointly normal with zero means, unit variances, $\mathrm{Corr}(\varepsilon, V) = 0.5$, and Z independent of (ε, V). Set $X = V+Z$ and $h_0(x) = \mathrm{sgn}(x-1)\log(1+|x-1|)$.

S and CNS. Let (X^*, Z^*, ε) be jointly standard normal with $\mathrm{Corr}(X^*, Z^*) = 0.5$, $\mathrm{Corr}(X^*, \varepsilon) = 0.3$, and $\mathrm{Corr}(Z^*, \varepsilon) = 0$. For S, set $X = 2\{F_N(X^*/3) - 1/2\}$, $Z = 2\{F_N(Z^*/3) - 1/2\}$, and $h_0(x) = 2\sin(\pi x)$. For CNS, set $X = F_N(X^*)$, $Z = F_N(Z^*)$, and $h_0(x) = 1 - 2F_N(x - 1/2)$.

CW. Let ζ, e, ν be independent standard normal variables. Set $Z = F_N(\zeta)$, $X = F_N(0.3\zeta + \sqrt{0.91}\, e)$, $\varepsilon = 0.5(0.3e + \sqrt{0.91}\, \nu)$, and $h_0(x) = 2(x - 1/2)_+^2 + x/2$.

CCK. Let (ε, V) be jointly standard normal with correlation 0.75. Independently, draw $Z^* \sim N(0, 1)$ and $D \sim \mathrm{Bernoulli}(1/2)$, independently of each other. Set $X = F_N(V + DZ^*)$, $Z = F_N(Z^*)$, and $h_0(x) = \sin(4x)\log x$. Since $X \in (0, 1)$ almost surely, this expression is well-defined.

A.2 Estimation and evaluation

For each model, a separate reference sample of 500,000 observations determines the marginal 1st and 99th percentiles of X and Z. These intervals are fixed across replications and sample sizes. Each constraint grid contains 101 empirical quantiles, at equally spaced probabilities between 0.01 and 0.99. The points need not be equally spaced in x or z. Training and test observations are not trimmed.

The structural basis is a cubic truncated-power spline with ten terms: a cubic polynomial and six truncated cubic terms at equally spaced interior knots on the reference interval for X. The same formulas are used to evaluate the spline outside this interval. The first stage is an unpenalized thin-plate regression spline of

dimension ten, including the constant, constructed from the training Z values using `mgcv::smoothCon` in R. We use the same basis and projection for the midpoint and QB within each replication. Let Φ and Ψ be the structural and first-stage design matrices. The common least-squares fit

$$\widehat{B} = (\Psi'\Psi)^{-1}\Psi'[Y, \Phi],$$

computed by QR decomposition, estimates g_0 and the conditional expectations of the structural basis functions. We evaluate the fitted regressions on the constraint grid using the same thin-plate basis. Thus the fitted residual for a spline h is $\widehat{g} - \widehat{\mathbb{A}}[h]$.

We restrict this residual to $[-b, b]$ at the 101 values of Z, and impose $|h''| \leq c$ at the 101 values of X. There is no range restriction. The combined matrix of moment and derivative constraints has full column rank in every replication, which ensures bounded pointwise envelopes whenever the constraints are feasible. No restrictions are imposed outside the reference intervals. Write

$$c = ac_0, \qquad a \in \{1/4, 1/2, 1, 1.5, 2, 4, 8\}.$$

For NP, S, CNS, CW, and CCK, respectively, we take

$$c_0 = 1, \quad 2\pi^2, \quad f_N(1/2), \quad 4, \quad 8.$$

The main text fixes $a = 1.5$. The other multipliers assess sensitivity to the restriction. For NP and CW, the reference values bound the second derivative wherever it exists; the first derivative is Lipschitz across the joining point, although the second derivative is undefined at $x = 1$ in NP and $x = 1/2$ in CW. For CCK, we use the fixed reference value $c_0 = 8$ instead of scaling by the large second derivative near zero. Its seven bounds are consequently $c \in \{2, 4, 8, 12, 16, 32, 64\}$. These restrictions serve as regularization choices and need not contain the true function. In particular, CCK's second derivative is unbounded as $x \downarrow 0$.

Using the residual quantiles defined in the estimation section, our initial tolerance for the correctly specified simulations is

$$b_{\text{initial}} = q_\eta(0.55) - q_\eta(0.45), \qquad \eta = Y - g_0(Z).$$

We evaluate the population distribution by Gaussian quadrature and numerically invert it at the two probabilities. This gives 0.334299, 0.456525, 0.239261, 0.157006, and 0.357327 for NP, S, CNS, CW, and CCK, respectively. These initial values are fixed across replications and sample sizes within each model. A feasible analogue would use empirical quantiles of $Y - \widehat{g}(Z)$; the present exercise uses the population quantities. After any unsuccessful attempt to compute all pointwise lower and upper bounds, the implementation multiplies b by 1.1 and repeats the calculation on the same sample, allowing at most 200 increases. This includes numerical failures and reported infeasibility. In the correctly specified simulations, all 70 combinations of model, multiplier, and sample size return 1000 successful replications. The misspecified simulations in Table 2 also retain all 1000 replications for each model and sample size.

For each training sample we draw 1000 test observations independently from the same model and evaluate the pointwise midpoints at their covariates. Training samples and test covariates are shared across the choices of a; test draws and reference grids are also shared across sample sizes. The reported loss includes observations outside the constraint intervals. The numerical stopping rule can make the final b depend on the test covariates through numerical failures; test outcomes are not used to choose b.

Writing $\ell_r = M^{-1}\sum_j(\widehat{h}_r^{\mathrm{mid}}(X'_{rj}) - h_0(X'_{rj}))^2$, we compute the Monte Carlo standard error of $\widehat{\mathcal{R}} = \sqrt{\bar{\ell}}$ as $s_\ell/(2\sqrt{R}\,\widehat{\mathcal{R}})$, where s_ℓ is the sample standard deviation of ℓ_r. We solve the linear programs using HiGHS (Huangfu and Hall, 2018). The underlying solver was checked against the original MATLAB results. For the specification reported here, we compared the computed envelopes with direct solutions of all individual linear programs in the replication with the largest loss in each of the 70 configurations.

A.3 Series and linear two-stage least squares

For series 2SLS, we follow the spline types and dimensions described by Kankanala (2025). The structural basis is a natural cubic spline of dimension J, including the constant, with $J-2$ interior knots at equally spaced empirical quantiles of the training X values. Its boundary knots are the training minimum and maximum; prediction is linear beyond these boundaries. The instrument basis is a thin-plate regression spline

of the same dimension, constructed from the training Z values. We use `splines::ns` and `mgcv::smoothCon` in R, with the default thin-plate construction, including its limit of 2000 distinct instrument values for constructing the basis. Both regressions are unpenalized.

If Φ_J and Ψ_J are the corresponding design matrices, the estimator has coefficients

$$\widehat{\beta}_J = (\Phi_J' P_{\Psi_J} \Phi_J)^{-1} \Phi_J' P_{\Psi_J} Y, \qquad P_{\Psi_J} = \Psi_J (\Psi_J' \Psi_J)^{-1} \Psi_J'.$$

We compute these coefficients after a QR decomposition of Ψ_J. The main table uses $J = 3$ at both sample sizes. Table 3 also reports $J = 4, 5, 6$ and the corresponding published $n = 1000$ values. Our estimates use the same training and test draws as the midpoint. All fits are retained, including those with very large errors. We checked the largest loss in each configuration by computing the two regression stages separately. The published description does not specify all basis-construction defaults, so our implementation need not reproduce its results exactly.

Linear 2SLS uses $\Phi = [\mathbf{1}, X]$ and $\Psi = [\mathbf{1}, Z]$. Equivalently, its estimated slope and intercept are

$$\widehat{\beta} = \frac{\sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})}{\sum_i (Z_i - \bar{Z})(X_i - \bar{X})}, \qquad \widehat{\alpha} = \bar{Y} - \widehat{\beta}\bar{X}.$$

We evaluate $\widehat{\alpha} + \widehat{\beta} X'$ on the same test draws. This estimator imposes a linear structural function, although the simulation functions are nonlinear. No replication is discarded.

To investigate the CW discrepancy at $J = 3$, we checked the model against the published specification and examined the first-stage matrix in the correctly specified simulations. In the replication with the largest error, the smallest canonical correlation between the structural and instrument bases is 0.000138, compared with a median of 0.0634. The nonlinear spline term is consequently very weakly identified in that sample. This replication accounts for 89.2% of the total squared error, and its fitted values range from -109 to 196, although $0 < h_0(X) < 1$. Computing the two regression stages separately reproduces the result. Ten additional independent batches, each with 1000 replications at $n = 1000$, give risks between 1.084 and 4.665; the median replication root mean squared error in each batch lies between 0.178 and 0.201. Thus the difference between 3.145 and the published 1.623 is consistent

with Monte Carlo variability from occasional large errors. Without the original implementation, we cannot exclude differences in spline construction as an additional explanation. Table 1 reports the initial batch, without discarding any replication.

A.4 Quasi-Bayes estimation

We implement the NPIV quasi-Bayes posterior mean described by Kankanala (2025). We standardize both X and Y using their training-sample means and standard deviations, and transform predictions back to the original units of Y. The first stage uses the same ten-dimensional thin-plate regression as the midpoint. We use identity weighting and a mean-zero Gaussian process with the conventional Matérn covariance of regularity 3/2,

$$C_{\sigma,\ell}(x,x') = \sigma^2 \left(1 + \frac{\sqrt{3}|x-x'|}{\ell}\right) \exp\left(-\frac{\sqrt{3}|x-x'|}{\ell}\right).$$

Independent LogNormal$(0,1)$ priors are placed on the amplitude σ and length scale ℓ, with no additional factor depending on K.[10]

For the linear NPIV residual, Gaussian integration permits direct calculation of the conditional posterior mean. To describe the calculation, write $\widetilde{\Psi}$ for an orthonormal basis of the first-stage regression space, C_θ for the covariance matrix on the standardized training covariates, and $S_\theta = I_K + \widetilde{\Psi}' C_\theta \widetilde{\Psi}$. Conditional on $\theta = (\sigma, \ell)$, the posterior mean at a standardized covariate x is

$$C_\theta(x, X)\widetilde{\Psi} S_\theta^{-1} \widetilde{\Psi}' Y,$$

where Y in this expression is standardized. Integrating out the function gives a hyperparameter likelihood proportional to $|S_\theta|^{-1/2} \exp\{-Y'\widetilde{\Psi} S_\theta^{-1} \widetilde{\Psi}' Y/2\}$. We combine this likelihood with the two lognormal priors, calculate the posterior means of σ and ℓ, and insert them into the conditional posterior mean above, as in the published procedure.

[10] The Fourier definition in equation (8) of Kankanala (2025) gives $(\pi/2)\sigma_F^2(1 + |x-x'|/\ell_F)$ $\exp(-|x-x'|/\ell_F)$ in one dimension. The two covariances coincide when $\sigma_F = \sqrt{2/\pi}\,\sigma$ and $\ell_F = \ell/\sqrt{3}$. Retaining identical lognormal priors under both parameterizations therefore changes the prior on functions. We specify our convention explicitly because we do not have the original simulation code.

Table 3: Series 2SLS risks and comparison with published results

Model	J	Published	This paper
NP	3	0.131	0.128
	4	0.154	0.154
	5	0.355	0.364
	6	4.840	29.119
S	3	0.292	0.293
	4	7.300	0.768
	5	37.520	19.073
	6	132.110	305.204
CNS	3	0.189	0.195
	4	11.770	13.856
	5	34.830	38.168
	6	74.350	51.967
CW	3	1.623	3.145
	4	8.200	18.568
	5	34.190	87.337
	6	113.370	49.056
CCK	3	0.345	0.349
	4	6.010	2.699
	5	130.040	10.266
	6	435.910	52.253

Notes: All entries are for $n = 1000$. Published values are from Table 1 of Kankanala (2025). Our calculations use $R = M = 1000$ and the same samples as the midpoint estimator. Both stages have dimension J, including the constant. No replication is discarded. Table 1 uses $J = 3$ in every model and at both sample sizes.

We calculate the hyperparameter means by numerical integration in their logarithms, refining the integration until both means change by less than 2×10^{-4} in relative terms. This avoids Markov chain simulation while targeting the same hyperparameter means. In one dimension, exact recursions for the Matérn covariance products also avoid factorizing an n-dimensional covariance matrix. We checked the predictions against direct Gaussian calculations and repeated the integration with finer spacing and wider bounds for the replications with the largest errors and largest integration changes. The true structural function is used only to evaluate predictions.

Table 4 compares our $n = 1000$ results with the published risks at $K = 5, 7, 10$, using the same prior in every case. These calculations are an independent implementation. Covariance parameterization and finite Markov chain error can affect the comparison. In diagnostic runs on 100 samples per model, 10,000-iteration exploration chains with acceptance rates near the published target sometimes give few effectively independent length-scale draws and materially different estimates from numerical integration. This does not establish the behavior of the original implementation. The reported estimates use numerical integration throughout.

Table 4: QB risks and comparison with published results

Model	K	Published	This paper
NP	5	0.155	0.147
	7	0.148	0.141
	10	0.141	0.135
S	5	0.232	0.206
	7	0.210	0.189
	10	0.197	0.177
CNS	5	0.138	0.139
	7	0.134	0.141
	10	0.134	0.145
CW	5	0.126	0.127
	7	0.122	0.124
	10	0.118	0.122
CCK	5	0.285	0.286
	7	0.276	0.274
	10	0.266	0.261

Notes: Root mean squared errors for NPIV QB at $n = 1000$. Table 1 uses $K = 10$. Published values are from Table 2 of Kankanala (2025); both sets of results use 1000 replications. The simulation draws and numerical implementation differ. The calculations in this paper use the covariance convention and standardization described above, and numerical integration for hyperparameter posterior means.

A.5 Sensitivity to the restrictions

Table 5 reports the midpoint risks at all seven multipliers and both sample sizes. The constraint grids and initial quantile calibration of b are the same as in Table 1. Bold entries correspond to $a = 1.5$.

Table 5: Sensitivity to the bound on the second derivative

Model	a	c	$n = 1000$	$n = 10{,}000$
NP	0.25	0.25	0.1346	0.0951
	0.5	0.5	0.1366	0.0875
	1	1	0.1456	0.0812
	1.5	1.5	**0.1550**	**0.0798**
	2	2	0.1630	0.0790
	4	4	0.1911	0.0778
	8	8	0.2388	0.0838
S	0.25	4.9348	0.2680	0.2436
	0.5	9.8696	0.2663	0.2414
	1	19.7392	0.2659	0.2347
	1.5	29.6088	**0.2666**	**0.2299**
	2	39.4784	0.2667	0.2252
	4	78.9568	0.2675	0.2065
	8	157.9137	0.2845	0.1699
CNS	0.25	0.088	0.1002	0.0408
	0.5	0.176	0.1003	0.0408
	1	0.3521	0.1005	0.0409
	1.5	0.5281	**0.1006**	**0.0409**
	2	0.7041	0.1009	0.0409
	4	1.4083	0.1024	0.0411
	8	2.8165	0.1073	0.0413
CW	0.25	1	0.1103	0.0836
	0.5	2	0.1096	0.0830
	1	4	0.1092	0.0820
	1.5	6	**0.1098**	**0.0808**
	2	8	0.1113	0.0791
	4	16	0.1230	0.0728
	8	32	0.1561	0.0688
CCK	0.25	2	0.3427	0.3465
	0.5	4	0.3337	0.3383
	1	8	0.3203	0.3232
	1.5	12	**0.3109**	**0.3104**
	2	16	0.3052	0.3002
	4	32	0.2949	0.2817
	8	64	0.2798	0.2557

Notes: Entries are the root mean squared errors in (5.1), with $R = M = 1000$. All specifications impose constraints at 101 points and use the same ten-dimensional thin-plate first stage, with no range restriction. The multipliers are $a = c/c_0$ for every model, with $c_0 = 8$ for CCK. Bold entries correspond to $a = 1.5$, the choice used in Table 1. Bounds c are rounded where necessary. Every specification starts from the same population 55th–45th percentile gap within each model.

B First-stage cross-validation

The cross-validation procedure uses a common ridge series fit for the response vector $(Y_i, \Phi_n(X_i)')'$. Candidate pairs consist of a first-stage dimension L and a ridge penalty λ. For each pair, compute mean held-out squared prediction error across folds, optionally scaling the response components to comparable units using training-fold scales. Let $\widehat{\mathrm{CV}}_{\min}$ be the smallest mean score and let $\widehat{\mathrm{SE}}_{\min}$ be the standard deviation of its fold scores divided by the square root of the number of folds. The one-standard-error rule retains pairs whose score is at most $\widehat{\mathrm{CV}}_{\min} + \widehat{\mathrm{SE}}_{\min}$, then chooses the smallest retained L and, among ties, the largest λ. This is a tuning heuristic, not a coverage or consistency theorem.

C Misspecification Diagnostics and the Falsification Frontier

Sensitivity analysis treats b as a user-specified index of tolerated misspecification. A complementary question is whether the data imply that a positive relaxation is required for the maintained model to be feasible. This is related to the moment-inequality diagnostics of Andrews and Kwon (2024). Here the diagnostic is the smallest index for which an admissible structural function exists.

C.1 Population falsification frontier

For a shape class $\mathcal{H}(c)$, define

$$\Theta_b(g_0; c) = \{h \in \mathcal{H}(c) : g_0 - \mathbb{A}[h] \in \mathcal{U}_b\}.$$

The minimum required relaxation, or falsification frontier, is

$$b^\star(c) = \inf\{b \geq 0 : \Theta_b(g_0; c) \neq \varnothing\}, \tag{C.1}$$

with $\inf \varnothing = +\infty$. For the leading supremum restriction, this reduces to

$$b^\star(c) = \inf_{h \in \mathcal{H}(c)} \|g_0 - \mathbb{A}[h]\|_\infty.$$

For the derivative restriction it instead minimizes $\|(g_0 - \mathbb{A}[h])'\|_\infty$ over $h \in \mathcal{H}(c)$. The normalization of the reported target does not enter this feasibility diagnostic.

By nesting, every $b > b^\star(c)$ is feasible when $b^\star(c) < \infty$, and every $b < b^\star(c)$ is infeasible. Feasibility at the boundary requires attainment. Sufficient conditions include compactness of the structural class and closedness of the joint feasibility restrictions in (h, b) in the relevant topology. For a fixed benchmark $b_0 \geq 0$, consider

$$H_0(b_0, c) : b^\star(c) \leq b_0, \qquad H_1(b_0, c) : b^\star(c) > b_0. \tag{C.2}$$

Under attainment, the null states feasibility at b_0; otherwise it includes a limiting boundary that need not itself be feasible.

Proposition C.1. *If $c_1 \leq c_2$ implies $\mathcal{H}(c_1) \subseteq \mathcal{H}(c_2)$, then $b^\star(c_2) \leq b^\star(c_1)$.*

Thus weaker shape restrictions cannot raise the minimum relaxation needed to reconcile the maintained model with the data.

C.2 Sample analog

Write $t(w)$ for the bound function when the scalar relaxation index is denoted t. Define

$$\begin{aligned} \hat{b}_n^\star(c) = \inf_{t \geq 0,\, \beta \in \mathbb{R}^{K_n}} \; & t \\ \text{s.t.} \quad & \mathbb{T}_U[\hat{g}_n - \hat{\Pi}_n'\beta](w) \leq t(w), && w \in \widetilde{W}_U, \\ & \mathbb{T}_H[\Phi_n'](x)\beta \leq C_c(x), && x \in \widetilde{S}_H. \end{aligned} \tag{C.3}$$

This is a linear program in (t, β) when $t(w)$ is affine in the scalar index t, as in both leading examples. For other nondecreasing bound functions, each fixed-t feasibility problem remains a linear program and the frontier can be found by a scalar feasibility search. Plotting $c \mapsto \hat{b}_n^\star(c)$ summarizes the trade-off between the restrictions on h and the relaxation needed for feasibility.

C.3 Consistency and a simple misspecification test

The rate for the index $b^\star$ also depends on how quickly the inequality bounds expand with that index. This is separate from the fixed-b endpoint problem. A sufficient

condition, on a fixed range $[0, B]$, is that for some $\lambda_U > 0$,

$$t_2(w) - t_1(w) \geq \lambda_U(t_2 - t_1)\mathbf{1}, \quad 0 \leq t_1 < t_2 \leq B, \quad w \in W_U. \tag{C.4}$$

Both the level and derivative examples have $\lambda_U = 1$. A nonlinear reindexing of $\mathcal{U}_b$ need not preserve a rate for $b^\star$, even if it preserves the collection of feasible sets.

Theorem C.2. *Fix c. Suppose Assumptions 3.1–3.3 and (3.3) hold for $\mathcal{H}(c)$, taking $\mathbb{L} = 0$ in Assumption 3.2 and with the grid bounds uniform on $[0, B]$. Suppose $\sup_{t\in[0,B]} \|t(\cdot)\|_\infty < \infty$, (C.4) holds, and $b^\star(c) < B < \infty$. Assume the shape anchor has uniformly positive moment slack at B: for some $\delta > 0$, $\mathbb{T}_U[g_0 - \mathbb{A}[h^s]](w) \leq B(w) - \delta\mathbf{1}$ for every $w \in W_U$. Here $h^s = 0$ under Assumption 3.2, or is the shape anchor in footnote 4 if that relaxation is used. Suppose also $\sup_{h\in\Theta_B} \|h\|_\infty < \infty$. Then $|\hat{b}_n^\star(c) - b^\star(c)| = O_p(r_n) = o_p(1)$, where $r_n = a_n + \kappa_n + d_n$.*

For the leading supremum restriction, it suffices to choose $B > 1 + \|g_0 - \mathbb{A}[h^s]\|_\infty$ and impose boundedness of Θ_B.

Corollary C.3. *Under Theorem C.2, let $\lambda_n \downarrow 0$ satisfy $r_n = o(\lambda_n)$. For fixed c and $b_0 \geq 0$, define*

$$\varphi_n(b_0, c) = \mathbf{1}\{\hat{b}_n^\star(c) > b_0 + \lambda_n\}.$$

Under $H_0(b_0, c)$, $P(\varphi_n = 1) \to 0$. Under each fixed alternative $b^\star(c) > b_0$, $P(\varphi_n = 1) \to 1$.

This test is pointwise consistent and conservative under the null. A nonzero asymptotic significance level requires additional distributional analysis. Like Andrews and Kwon (2024), the diagnostic addresses possible incompatibility between the restrictions and the observable law. Here the frontier specifically measures the relaxation needed within an NPIV model and a chosen structural shape class.

D Primitive conditions for the first stage and sieve

For the series estimates in (2.1)–(2.2), write $Q_n = E[\Psi_n(Z)\Psi_n(Z)']$.

For $s > 0$, let $m_s = \lceil s \rceil - 1$ and $\delta_s = s - m_s \in (0, 1]$. The Hölder norm $\|f\|_{C^s}$ bounds all partial derivatives through order m_s, including the function itself, and the δ_s-Hölder seminorms of derivatives of order m_s. This convention includes Lipschitz functions at $s = 1$.

Assumption D.1 (Design and conditional smoothness)**.** The eigenvalues of Q_n are uniformly bounded above and away from zero, and S_Z is a bounded subset of $\mathbb{R}^{d_Z}$. There is a conditional density $f_{X|Z}$ with respect to a measure μ_X and an integrable envelope $\ell(x)$ such that

$$\|f_{X|Z}(x \mid \cdot)\|_{C^s} \leq \ell(x), \qquad \int \ell(x)\, d\mu_X(x) < \infty.$$

Assumption D.2 (Series approximation and entropy)**.** Define the population series projection by

$$P_n f(z) = \Psi_n(z)' Q_n^{-1} E[\Psi_n(Z) f(Z)].$$

There is $R_n(s) \downarrow 0$ such that $\|f - P_n f\|_\infty \leq \|f\|_{C^s} R_n(s)$. Also $\sup_z \|\Psi_n(z)\|_2 \leq \bar{\xi}_n$, $\Psi_n(z) \neq 0$, and $\alpha_n(z) = \Psi_n(z)/\|\Psi_n(z)\|_2$ is Lipschitz with constant $\ell_n \geq 1$. Finally,

$$\int_0^1 \sqrt{\log N(\mathcal{H}, \|\cdot\|_\infty, u)}\, du < \infty, \qquad \frac{\bar{\xi}_n^2 \log(2L_n)}{n} \longrightarrow 0.$$

Here N denotes a covering number in the stated norm.

The entropy condition implies that $\mathcal{H}$ is totally bounded, and therefore uniformly bounded in supremum norm. Thus the first-stage argument does not require a separate condition that the shape restrictions imply pointwise bounds. The integrable envelope in Assumption D.1 justifies differentiation under the integral and bounds the smoothness of $\mathbb{A}[h]$ by a constant times $\|h\|_\infty$.

Assumption D.3 (Reduced-form moments and growth)**.** The function g_0 has finite C^s norm. For some $m > 2$,

$$\sup_{z \in S_Z} E[|Y - g_0(Z)|^m \mid Z = z] < \infty.$$

For some $v > 0$, as $L_n \to \infty$,

$$\bar{\xi}_n = O(\sqrt{L_n}), \quad \log \ell_n = O(\log L_n), \quad R_n(s) = O(L_n^{-v/d_Z}), \quad R_n(s) = O(L_n^{-1/2}),$$

and

$$\frac{\bar{\xi}_n^{2m/(m-2)} \log(2L_n)}{n} = O(1), \qquad \frac{L_n \log(2L_n)}{n^{1-2/m}} = O(1).$$

Theorem D.1 (First stage for the leading supremum restriction)**.** *Under Assumptions D.1–D.3, the estimates in (2.1)–(2.2), with* $\mathcal{U}_b = \{u : \|u\|_\infty \leq b\}$*, satisfy Assumption 3.1 with*

$$a_n = \sqrt{L_n \log(2L_n)/n} + L_n^{-v/d_Z}.$$

The bound is uniform over all structural sieve dimensions K_n *and bases whose admissible functions belong to the fixed class* $\mathcal{H}$*.*

Balancing the two displayed terms gives $L_n \asymp (n/\log n)^{d_Z/(2v+d_Z)}$ and $a_n \asymp (\log n/n)^{v/(2v+d_Z)}$, provided this choice satisfies all the moment and growth conditions. The theorem concerns least squares first stages. For ridge first stages, ridge bias must also be included in a_n unless it is negligible at this rate.

Proposition D.2 (A sufficient transformed first-stage rate)**.**

Suppose Assumptions D.1 and D.2 hold, $g_0 \in C^s$*, and* $\sup_z E[(Y - g_0(Z))^2 \mid Z = z] < \infty$*. Suppose the fixed operator* $\mathbb{T}_U$ *is defined on the relevant* C^s *functions and the first-stage sieve, and*

$$\|\mathbb{T}_U[f - P_n f]\|_\infty \leq \|f\|_{C^s} R_{U,n}(s), \qquad R_{U,n}(s) \to 0.$$

Put $\zeta_{U,n} = \sup_{w,j} \|(\mathbb{T}_U[\Psi_n'](w))_{j,\cdot}\|_2$*. For the series estimates, a sufficient rate in Assumption 3.1 is*

$$a_n = \zeta_{U,n}\sqrt{L_n/n} + R_{U,n}(s), \tag{D.1}$$

provided the right side tends to zero. The bound is uniform over structural sieve dimensions whose admissible functions lie in $\mathcal{H}$*.*

This sufficient bound is conservative; the preceding theorem gives the sharper rate for the leading supremum restriction. A known operator bounded in supremum norm inherits that sharper rate by its operator norm. A derivative operator is not bounded in that norm: its transformed basis size $\zeta_{U,n}$ and projection error $R_{U,n}(s)$ must be controlled separately. When $s > 1$, the conditional-density envelope in Assumption D.1 also gives (3.3) for first-derivative constraints.

Theorem D.3 (Range- and curvature-preserving splines)**.** *Let* $S_X = [x_-, x_+]$ *be a nondegenerate compact interval,* $c > 0$*, and*

$$\mathcal{H}(c) = \{h \in C^2(S_X) : 0 \leq h \leq 1,\ \|h''\|_\infty \leq c\}.$$

Let Φ_n span the splines of a fixed order $q \geq 4$ with equally spaced simple interior knots and mesh width $\Delta_n \asymp K_n^{-1}$, where K_n is the dimension of the spline space. There are $q_n[h] \in \mathcal{H}(c) \cap \operatorname{span}(\Phi_n)$ such that

$$\sup_{h \in \mathcal{H}(c)} \left\{ \|q_n[h] - h\|_\infty + \|(q_n[h])' - h'\|_\infty \right\} = O(K_n^{-1/2}).$$

Thus Assumption 3.2, with the relaxation in footnote 4, holds with $\kappa_n = O(K_n^{-1/2})$ for point evaluation, centered evaluation, derivative evaluation, and average derivatives $\mathbb{L}[h] = E[h'(X)]$, using $h^s = q_n^s = 1/2$. The class has compact closure in the supremum norm and satisfies the entropy condition in Assumption D.2. Point and average first derivatives extend continuously to this closure, with derivative evaluations equicontinuous over the evaluation point. For the average derivative, $\widehat{\mathbb{L}}_n[h] = n^{-1} \sum_i h'(X_i)$ satisfies Assumption 3.4 with $a_{\mathbb{L},n} = n^{-1/2}$.

The order convention is degree plus one, so the theorem includes cubic splines. The curvature operators applied to this basis are continuous and piecewise polynomial, hence Lipschitz on the compact interval, as required for the grid comparison.

E Proofs

Proof of Proposition 3.1. Put $d = \|(b - \epsilon)^{-1}\|_\infty^{-1} > \eta$ and $c^* = \mathbb{L}[h^*]$. Every component of $b(w) - \epsilon(w)$ is at least d, so h^* has population moment slack at least d and belongs to both feasible sets. If $\tilde{h}$ is perturbed-feasible, every population inequality is violated by at most η. Hence

$$h^\circ = \frac{d}{d+\eta}\tilde{h} + \frac{\eta}{d+\eta}h^*$$

is population-feasible: convexity of $\mathcal{H}$ preserves its shape restrictions, and the excess in each moment inequality is at most $\{d\eta - \eta d\}/(d+\eta) = 0$. Linearity of $\mathbb{L}$ gives

$$\mathbb{L}[\tilde{h}] - \bar{\theta}_{\mathbb{L}}(b) \leq \frac{\eta}{d}\{\bar{\theta}_{\mathbb{L}}(b) - c^*\}.$$

Applying this argument to $-\mathbb{L}$ proves finiteness and both outward bounds. Conversely, if h is population-feasible, then $(1 - \eta/d)h + (\eta/d)h^*$ has population moment slack at least η. Its perturbation error is at most η, so it is perturbed-feasible. Taking

suprema gives the inward upper bound; replacing $\mathbb{L}$ by $-\mathbb{L}$ gives the lower bound. Average the endpoint bounds for the midpoint. No optimizer need exist. ∎

Proof of Theorem 3.2. We use the relaxed shape condition in footnote 4. In addition to the feasible approximations $q_n[h]$, take $h^s \in \mathcal{H}$, $q_n^s \in \mathcal{H}_n$, and $\sigma > 0$ with

$$\|q_n^s - h^s\|_\infty \le \kappa_n, \qquad \mathbb{T}_H[q_n^s](x) \le C(x) - \sigma \mathbf{1}, \quad x \in S_H.$$

Assumption 3.2 implies this condition with $h^s = q_n^s = 0$: zero belongs to every linear sieve and $\mathbb{T}_H[0] = 0 \le C(x) - \sigma\mathbf{1}$. The argument below allows the general h^s and q_n^s, establishing the stated relaxation.

Constants below are independent of n, b, and the functional in the stated ranges. We use $q_n[h]$ only for $h \in \Theta_b \subseteq \Theta_{\bar{b}}$, including h^*. Since these functions have norm at most M, the approximation bounds in Assumption 3.2 apply on a fixed ball, with constants independent of b. Work on events of probability approaching one where the grid bounds hold and the errors divided by $s(\underline{b}_n)$ are small. We repeatedly use one repair: if a candidate violates each affine inequality by at most $v \ge 0$ and an anchor has slack $\mu > 0$, mixing with weight $v/(\mu+v)$ on the anchor satisfies every inequality.

A common feasible anchor. Every component of $b(w) - \epsilon(w)$ is at least $s(b)$, so the fixed h^* has moment slack at least $s(b)$ throughout the stated index range. Since $h^* \in \Theta_{\bar{b}}$, its norm is at most M. Set

$$D = 1 + C_A(\|h^s\|_\infty + M), \qquad t_b = \frac{s(b)}{2D}, \qquad q_{n,b} = (1 - t_b)q_n[h^*] + t_b q_n^s.$$

Because $s(b) \le 1$ and $D \ge 1$, $0 < t_b \le 1/2$. The unsieved mixture has moment slack at least $s(b) - t_b D = s(b)/2$. By (3.3), $q_{n,b}$ has moment slack at least $s(b)/2 - C\kappa_n$ and shape slack at least $t_b\sigma$. Its norm is at most a fixed A_0. Since $q_{n,b} \in \mathcal{H}_n$, its empirical moment error is at most η_n. For some fixed $c_0 > 0$, its population and continuum empirical inequalities therefore all have slack at least $\mu_b = c_0 s(b)$, uniformly in b, on the stated events. This proves nonemptiness of all the feasible sets below.

Population sieve approximation. Let $F_b^* = \Theta_b \cap \operatorname{span}(\Phi_n)$. For $h \in \Theta_b$, the approximation $q_n[h] \in \mathcal{H}_n$ violates each population moment inequality by at most $C\kappa_n$. Mix it toward $q_{n,b}$. The resulting function is in F_b^* and is within $C\kappa_n/s(b)$ of h in supremum norm, using $\|h\|_\infty \le M$. The reverse inclusion is immediate.

First-stage perturbation. Let

$$F_b^\circ = \{h = \Phi_n'\beta \in \mathcal{H}_n : \mathbb{T}_U[\hat{g}_n - \hat{\Pi}_n'\beta] \le b(\cdot)\}.$$

If $h \in F_b^\circ$, its population moment violation is at most η_n. Repair toward $q_{n,b}$ to obtain an element of F_b^*. Solving the mixture identity yields

$$\|h\|_\infty \le (1 + \eta_n/\mu_b)M + (\eta_n/\mu_b)A_0.$$

This is bounded by a fixed M_1, simultaneously in b. The same repair in the opposite direction maps F_b^* into F_b°. Both distances are at most $C\eta_n/s(b)$. The first-stage bound has been applied only to functions in $\mathcal{H}_n$.

Finite grids and boundedness. Let F_b^g be the actual grid-feasible set. For $h = \Phi_n'\beta \in F_b^g$, the maximum violation of a continuum empirical inequality is at most

$$v_n(h) \le \{\xi_n D_{H,n} + 2G_n D_{U,n}\}(1 + \|\beta\|_2) \le 2d_n(1 + \|h\|_\infty).$$

Mixing toward $q_{n,b}$ gives an element of F_b° and hence

$$\|h\|_\infty \le M_1 + \frac{2d_n(M_1 + A_0)}{\mu_b}(1 + \|h\|_\infty).$$

The coefficient is eventually at most $1/2$, uniformly in b. Every grid-feasible function therefore has norm at most $2M_1 + 1$, and its distance from F_b° is at most $Cd_n/s(b)$. The reverse inclusion is immediate.

Functional values and estimation. All shape-feasible functions used above lie in a fixed supremum-norm ball. Assumption 3.4 therefore bounds their $\mathbb{L}$ values uniformly. For a repair $u = (1-t)h + tq_{n,b}$ with $h \in \mathcal{H}$, linearity gives

$$|\mathbb{L}[u] - \mathbb{L}[h]| = t|\mathbb{L}[q_{n,b}] - \mathbb{L}[h]| \le Ct.$$

Assumption 3.2 also bounds $|\mathbb{L}[q_n[h]] - \mathbb{L}[h]|$ by $C\kappa_n$ for $h \in \Theta_b$. Thus the population-sieve comparison changes functional values by at most $C\kappa_n/s(b)$, and the first-stage repairs change them by at most $C\eta_n/s(b)$.

For a grid-feasible h, its repair $u = (1-t)h + tq_{n,b}$ belongs to F_b° and has $t/(1-t) = v_n(h)/\mu_b \le Cd_n/s(b)$. Although h need not belong to $\mathcal{H}$, solving the same identity

gives

$$\mathbb{L}[h] - \mathbb{L}[u] = \frac{t}{1-t}\{\mathbb{L}[u] - \mathbb{L}[q_{n,b}]\}.$$

The right side is bounded in absolute value by $Cd_n/s(b)$. Combining the three comparisons gives, in both directions between Θ_b and F_b^g, pairs whose functional values differ by at most $C(\kappa_n + \eta_n + d_n)/s(b)$.

The estimation-error bound applies uniformly to u and $q_{n,b}$, which lie in a fixed supremum-norm ball in $\mathcal{H}_n$. Linearity also gives

$$(\widehat{\mathbb{L}}_n - \mathbb{L})[h] = \frac{(\widehat{\mathbb{L}}_n - \mathbb{L})[u] - t(\widehat{\mathbb{L}}_n - \mathbb{L})[q_{n,b}]}{1-t}.$$

Since $t = o(1)$ uniformly on the stated events, the estimation error is $O_p(a_{\mathbb{L},n})$ uniformly over all grid-feasible functions and b in the stated range. Adding this error to the functional-value comparisons and taking suprema or infima proves (3.6). All bounds are uniform over families satisfying the stated uniform assumptions. Averaging gives the midpoint result. Assigning arbitrary finite values on the exceptional event of infeasibility does not affect these probability statements. ∎

Proof of Corollary 3.3. If $\omega_{\mathbb{L}}(b)$ did not tend to zero, there would be $b_j \downarrow 0$ and $h_j \in \Theta_{b_j}$ whose target values stay a positive distance from $\mathbb{L}[h_0]$. Compactness gives a uniformly convergent subsequence with limit $h_\infty \in \overline{\mathcal{H}}$. Continuity of $\mathbb{T}_U\mathbb{A}$, together with pointwise convergence of the bounds as the scalar index tends to zero, implies that h_∞ satisfies the moment inequalities at index zero. Identification at zero tolerance gives $\mathbb{L}[h_\infty] = \mathbb{L}[h_0]$, and continuity of the extension then gives a contradiction. For a family of functionals, equicontinuity makes the difference between their values at h_j and h_∞ tend to zero uniformly, while identification makes the difference between their values at h_∞ and h_0 zero.

Both endpoints of the target interval lie within $\omega_{\mathbb{L}}(b)$ of $\mathbb{L}[h_0]$, as does their midpoint. Apply Theorem 3.2 with $\underline{b}_n = b_n$ and the triangle inequality. Balancing b_n^α and r_n/b_n^ν gives the power rate and satisfies $r_n = o(s(b_n))$; the objective error adds $a_{\mathbb{L},n}$. ∎

Theorem E.1 (Several conditional-moment restrictions)**.** *For a fixed finite collection of instrument vectors, impose their transformed conditional-moment inequalities jointly. Suppose the assumptions of Theorem 3.2 hold for the stacked operators and grids, with maximum first-stage error $O_p(a_n)$ over the fixed number of instrument*

vectors. Then the same rates hold with $s(b)$ computed from the stacked functions b and fixed ϵ, including the additive error for estimated functionals.

Proof. Stacking finitely many affine constraint systems preserves convexity and linearity. Take maxima of their first-stage and grid errors and the minimum componentwise gap across the systems. Every repair and objective comparison in the preceding theorem applies to the stacked system. ∎

Lemma E.2 (Uniform series first stage)**.** *Under Assumptions D.1 and D.2, put $\hat{\gamma}[h] = \hat{Q}_n^{-1} n^{-1} \sum_i \Psi_n(Z_i) h(X_i)$ on the nonsingular event. Then*

$$\sup_{h \in \mathcal{H}} \|\Psi_n' \hat{\gamma}[h] - \mathbb{A}[h]\|_\infty = O_p\left(\frac{\bar{\xi}_n}{\sqrt{n}} \{1 + \sqrt{\log(2\ell_n)} + \sqrt{L_n} R_n(s)\} + R_n(s) \right).$$

This bound is $o_p(1)$ whenever its displayed deterministic rate tends to zero.

Proof. The entropy condition makes $\mathcal{H}$ totally bounded. In particular,

$$M_H := \sup_{h \in \mathcal{H}} \|h\|_\infty < \infty.$$

Indeed, finiteness of the integral ensures a finite cover at some positive radius, which already gives boundedness; monotonicity then ensures finite covers at every positive radius. Write $C_A = \int \ell \, d\mu_X$. Differentiating under the integral, and integrating the Hölder difference bounds, gives

$$\|\mathbb{A}[h]\|_{C^s} \le C_A \|h\|_\infty.$$

This also holds for differences of two functions in $\mathcal{H}$.

Whiten Ψ_n by $Q_n^{-1/2}$. Uniform eigenvalue bounds change the bounds on its norm and its normalized-direction Lipschitz constant only by fixed factors, and do not change the series projection or fitted values. We can therefore suppose $Q_n = I$. Let $\gamma[h] = E[\Psi_n(Z) h(X)]$, and define

$$e_i[h] = h(X_i) - \mathbb{A}[h](Z_i), \qquad v_h(z) = \mathbb{A}[h](z) - \Psi_n(z)' \gamma[h].$$

Then $|e_i[h]| \le 2M_H$, $|e_i[h] - e_i[k]| \le 2\|h - k\|_\infty$, and

$$\|v_h\|_\infty \le C_A M_H R_n(s), \qquad \|v_h - v_k\|_\infty \le C_A R_n(s) \|h - k\|_\infty. \tag{E.1}$$

The matrix law of large numbers (Rudelson, 1999; Belloni et al., 2015, Lemma 6.2) gives $\|\hat{Q}_n - I\|_{\text{op}} = o_p(1)$. In what follows we restrict to the event where its eigenvalues lie in $[1/2, 2]$. All constants in the conditional bounds on this event are deterministic.

The least squares decomposition is

$$\sqrt{n}\,\alpha_n(z)'(\hat{\gamma}[h] - \gamma[h]) = \alpha_n(z)'\hat{Q}_n^{-1} n^{-1/2} \sum_i \Psi_n(Z_i) e_i[h] + \alpha_n(z)'\hat{Q}_n^{-1} n^{-1/2} \sum_i \Psi_n(Z_i) v_h(Z_i).$$

For the first process, condition on $Z_1, \ldots, Z_n$ and symmetrize the independent, conditionally mean-zero errors using independent Rademacher variables ϵ_i. The empirical L^2 distance between the coefficient arrays indexed by (z, h) and (z', k) is bounded by

$$C\{M_H \ell_n \|z - z'\|_2 + \|h - k\|_\infty\}.$$

Split the difference into its z and h components and use

$$n^{-1} \sum_i \Psi_n(Z_i)\Psi_n(Z_i)' = \hat{Q}_n$$

to obtain this metric bound. Adjoin the zero array and the negatives of the arrays when bounding the absolute supremum; this adds only a constant to the entropy bound. The arrays have bounded diameter. A product cover for S_Z and $\mathcal{H}$, the bound $N(S_Z, \|\cdot\|_2, u) \le C(1 + 1/u)^{d_Z}$, and the entropy integral for $\mathcal{H}$ show that their Dudley integral is at most $C\{1 + \sqrt{\log(2\ell_n)}\}$. Conditional symmetrization, Dudley's inequality (Dudley, 1967), and Markov's inequality thus give

$$\sup_{z,h} \left| \alpha_n(z)'\hat{Q}_n^{-1} n^{-1/2} \sum_i \Psi_n(Z_i) e_i[h] \right| = O_p(1 + \sqrt{\log(2\ell_n)}).$$

Restricting to the well-conditioned event before taking conditional expectations avoids any moment requirement on $\hat{Q}_n^{-1}$.

For the second process, use projection orthogonality $E[\Psi_n(Z) v_h(Z)] = 0$ and bound its norm by the supremum over $\|u\|_2 \le 1$ of $|n^{-1/2} \sum_i u'\Psi_n(Z_i) v_h(Z_i)|$, times

$\|\hat{Q}_n^{-1}\|_{\text{op}}$. After symmetrization, its empirical metric is bounded by

$$CR_n(s)\|\hat{Q}_n^{1/2}\|_{\text{op}}\{M_H\|u-u'\|_2+\|h-k\|_\infty\}.$$

Use (E.1) and $N(\{u : \|u\|_2 \le 1\}, \|\cdot\|_2, t) \le (1+2/t)^{L_n}$. The resulting entropy integral is bounded by $CR_n(s)\|\hat{Q}_n^{1/2}\|_{\text{op}}(1+\sqrt{L_n})$. The matrix expectation bound also implies $E\|\hat{Q}_n^{1/2}\|_{\text{op}} = O(1)$, so symmetrization and Markov give $O_p(\sqrt{L_n}R_n(s))$ for this process. Multiplying by $\bar{\xi}_n/\sqrt{n}$ and adding the projection bias $C_A M_H R_n(s)$ proves the lemma. ∎

Proof of Theorem D.1. For $h = \Phi_n'\beta \in \mathcal{H}_n$, $\hat{\Pi}_n'\beta = \Psi_n'\hat{\gamma}[h]$ and $\Pi_n'\beta = \mathbb{A}[h]$. The preceding lemma bounds the error uniformly over the larger, fixed class $\mathcal{H}$. Since $\sqrt{L_n}R_n(s) = O(1)$ and $\log \ell_n = O(\log L_n)$, its bound is $O_p(a_n)$ and contains no K_n.

For $\hat{g}_n$, apply Belloni et al. (2015), Theorem 4.3. Their conditional error-moment requirement is supplied by the uniform bound on $E[|Y - g_0(Z)|^m \mid Z]$. Use their approximation bound $\ell_k c_k = O(R_n(s))$. After division by $\sqrt{\log(2L_n)}$, the two remainder bounds have orders

$$\sqrt{\frac{L_n \log(2L_n)}{n^{1-2/m}}} + \frac{L_n R_n(s)}{\sqrt{n}}, \qquad R_n(s),$$

respectively. These are bounded under the stated assumptions. Their design, basis, and sample conditions are also satisfied. Consequently $\|\hat{g}_n - g_0\|_\infty = O_p(a_n)$. Adding the two errors proves the claim. Since $L_n \log(2L_n)/n \to 0$ and $L_n^{-v/d_Z} \to 0$, this rate vanishes. ∎

Proof of Proposition D.2. The density envelope gives $\|\mathbb{A}[h]\|_{C^s} \le C_A\|h\|_\infty$, also for differences. The entropy condition bounds $\mathcal{H}$ in supremum norm by M_H. Write $\gamma[h] = Q_n^{-1}E[\Psi_n(Z)h(X)]$ and $\hat{\gamma}[h] = \hat{Q}_n^{-1}n^{-1}\sum_i \Psi_n(Z_i)h(X_i)$. On the event where $\hat{Q}_n$ has eigenvalues bounded away from zero, the least-squares identity gives

$$\hat{\gamma}[h] - \gamma[h] = \hat{Q}_n^{-1}\frac{1}{n}\sum_i \Psi_n(Z_i)\{h(X_i) - P_n\mathbb{A}[h](Z_i)\}.$$

The summands have mean zero by projection orthogonality. Moreover, $r_h(X,Z) = h(X) - P_n\mathbb{A}[h](Z)$ is uniformly bounded, and $\|r_h - r_k\|_\infty \le C\|h-k\|_\infty$, since the untransformed projection error is at most $C_A R_n(s)\|h-k\|_\infty$.

Bound the norm of the sum by the supremum over $\|v\|_2 \le 1$ and $h \in \mathcal{H}$ of its scalar products. After symmetrization, the empirical metric of these coefficient arrays is at most

$$C\|\hat{Q}_n^{1/2}\|_{\mathrm{op}}\{M_H\|v - v'\|_2 + \|h - k\|_\infty\}.$$

The Euclidean unit ball has covering number at most $(1 + 2/t)^{L_n}$, and the entropy integral of $\mathcal{H}$ is finite. Their product cover gives a Dudley bound of order $\|\hat{Q}_n^{1/2}\|_{\mathrm{op}}(\sqrt{L_n} + 1)$ for the $n^{-1/2}$-scaled process. The same matrix bound used in Lemma E.2 implies $E\|\hat{Q}_n^{1/2}\|_{\mathrm{op}} = O(1)$ and boundedness in probability of $\hat{Q}_n^{-1}$. Consequently

$$\sup_{h\in\mathcal{H}} \|\hat{\gamma}[h] - \gamma[h]\|_2 = O_p(\sqrt{L_n/n}).$$

For the reduced-form coefficient, the centered score is $\Psi_n(Z)\{Y - P_n g_0(Z)\}$. Its squared norm has expectation $O(L_n)$ by the conditional variance bound, the bounded projection bias, and $E\|\Psi_n(Z)\|_2^2 = \mathrm{tr}(Q_n) = O(L_n)$. The same coefficient rate follows by Markov's inequality.

Applying $\mathbb{T}_U$ to the fitted functions multiplies these coefficient errors by at most $\zeta_{U,n}$. Their transformed population projection biases are bounded by $CR_{U,n}(s)$. This proves (D.1). ∎

Proof of Theorem D.3. Put $I = [x_-, x_+]$ and $L = x_+ - x_- > 0$. Taylor's formula, evaluated from any point toward the more distant endpoint, gives the common derivative bound $\|h'\|_\infty \le 2/L + cL =: B_1$ for $h \in \mathcal{H}(c)$. Extend h outside I by its endpoint tangent lines, obtaining H with a globally Lipschitz derivative, $\|H'\|_\infty \le B_1$, and $\|H''\|_\infty \le c$ in the weak-derivative sense. Unlike a constant extension, this extension has no jumps in its first derivative at the endpoints.

For $\rho > 0$, define $H_\rho(x) = (2\rho)^{-1}\int_{-\rho}^{\rho} H(x + u)\,du$. Then $\|H_\rho - h\|_{\infty,I} \le B_1\rho$ and $\|H_\rho' - h'\|_{\infty,I} \le c\rho$. Also,

$$H_\rho''(x) = \frac{H'(x + \rho) - H'(x - \rho)}{2\rho}$$

has supremum norm at most c and Lipschitz constant at most c/ρ. Use the nonnegative, partition-of-unity B-spline basis $\{B_{j,q-2}\}$ of order $q - 2$ on the same mesh to form

$$w_n(x) = \sum_j H_\rho''(\zeta_j)B_{j,q-2}(x),$$

where ζ_j is any point in the support of the corresponding basis function. Each such support has length at most $C_q\Delta_n$. Therefore

$$\|w_n\|_\infty \le c, \qquad \|w_n - H_\rho''\|_\infty \le C_q c\Delta_n/\rho.$$

These elementary basis properties are described in de Boor (2014). Integrate w_n twice, choosing the integration constants to agree with H_ρ and H_ρ' at x_-. The resulting function p_n is an order-q spline in the original sieve, with $\|p_n''\|_\infty \le c$, and

$$\|p_n - h\|_\infty \le B_1\rho + (L^2/2)C_q c\Delta_n/\rho.$$

Similarly, integrating the second-derivative approximation once gives

$$\|p_n' - h'\|_\infty \le c\rho + LC_q c\Delta_n/\rho.$$

Take $\rho = \sqrt{\Delta_n}$ and denote the bound on $\|p_n - h\|_\infty$ by $e_n = O(\sqrt{\Delta_n})$. Since $-e_n \le p_n \le 1 + e_n$, the spline

$$q_n[h] = \frac{p_n + e_n}{1 + 2e_n}$$

lies in $[0,1]$, has curvature bounded by c, and obeys $\|q_n[h] - h\|_\infty \le 2e_n$. Moreover,

$$\|(q_n[h])' - h'\|_\infty \le \|p_n' - h'\|_\infty + 2e_n B_1 = O(\sqrt{\Delta_n}),$$

uniformly in h. This proves the joint approximation and its implications for the stated functionals. For $q \ge 4$ these splines are C^2, so they belong to $\mathcal{H}(c)$. The constant spline $1/2$ has common shape slack at least $\min\{1/2, c\}$ for the restrictions $(h, -h, h'', -h'')' \le (1, 0, c, c)'$.

The common value and derivative bounds make $\mathcal{H}(c)$ uniformly bounded and equicontinuous. Arzelà–Ascoli gives compact closure in $C(I)$. To check the entropy integral directly, choose a mesh of size proportional to u and round the values of h at its nodes to a grid of size proportional to u. Linear interpolation gives a supremum-norm cover with error at most u, after adjusting the fixed constants. There are $O(u^{-1})$ nodes and $O(u^{-1})$ possible rounded values at each node, so $\log N(\mathcal{H}(c), \|\cdot\|_\infty, u) \le Cu^{-1}\log(C/u)$ for $0 < u < 1$. The square root of this bound is integrable at zero.

For $h, k \in \mathcal{H}(c)$, put $\delta = \|h - k\|_\infty$. Taylor's formula toward the more distant endpoint gives, for $0 < t \le L/2$,

$$\|h' - k'\|_\infty \le 2\delta/t + ct.$$

Choosing t of order $\sqrt{\delta}$ for small δ proves uniform continuity of derivative evaluation, uniformly in its evaluation point. If a sequence in $\mathcal{H}(c)$ converges uniformly, its derivatives therefore converge uniformly too, and their limit is the derivative of the limiting function. This defines the continuous extensions to $\overline{\mathcal{H}(c)}$, including average derivatives.

Finally, the class $\{h' : h \in \mathcal{H}(c)\}$ is uniformly bounded by B_1 and has common Lipschitz constant c. The same mesh-and-rounding argument gives it a finite supremum-norm entropy integral. Symmetrization and Dudley's inequality, as in Lemma E.2, yield

$$\sup_{h \in \mathcal{H}(c)} \left| \frac{1}{n} \sum_i h'(X_i) - E[h'(X)] \right| = O_p(n^{-1/2}).$$

Together with $|E[h'(X)]| \le B_1$, this verifies Assumption 3.4 for the average derivative. ∎

E.1 Proofs for Appendix C

Proof of Proposition C.1. Every feasible pair (b, h) for $\mathcal{H}(c_1)$ is feasible for $\mathcal{H}(c_2)$. Taking the infimum over the latter set cannot increase the minimum relaxation. ∎

Proof of Theorem C.2. Suppress c and work on the grid and first-stage events used in the proof of Theorem 3.2, using the shape anchors h^s and q_n^s from that proof. The latter has fixed positive shape slack and population moment slack at least $\delta - C_A\kappa_n$ at B. Its empirical moment slack is at least $\delta - C_A\kappa_n - \eta_n$, and hence at least $\delta/2$ with probability approaching one. Its transformed residuals are bounded uniformly over the index range $[0, B]$.

For $\epsilon > 0$ with $b^\star + \epsilon < B$, take $h \in \Theta_{b^\star+\epsilon} \subseteq \Theta_B$. Boundedness of Θ_B makes the approximation bounds in Assumption 3.2 uniform in ϵ. The approximation $q_n[h]$ is shape-feasible and has empirical moment violation at most $C\kappa_n + \eta_n$. Increasing the index by $(C\kappa_n + \eta_n)/\lambda_U$ repairs those inequalities by (C.4). The function is feasible

on both grids. Letting $\epsilon \downarrow 0$ gives

$$\hat{b}_n^\star \leq b^\star + (C\kappa_n + \eta_n)/\lambda_U.$$

Thus the sample frontier is below a fixed number less than B with probability approaching one.

The boundedness argument of Theorem 3.2, at the single index B with moment anchor h^s, bounds every grid-feasible function at B by a fixed constant. By nesting this includes every candidate at an index $t \leq B$. Take such a candidate with index arbitrarily close to $\hat{b}_n^\star$. Its continuum empirical inequality violations are $O(d_n)$. Mix toward q_n^s with weight $O(d_n)$ to repair the shape constraints. By affinity, each repaired moment violation is bounded above by the candidate's $O(d_n)$ violation plus $O(d_n)$ times the anchor's violation. The latter is bounded uniformly in t because the anchor residual and the bound functions $t(\cdot)$ are bounded. Thus the repaired moment violation is $O(d_n)$. The result belongs to $\mathcal{H}_n$, so its population moment violation is $O(d_n) + \eta_n$ by the first-stage bound. Increasing t by $C(d_n + \eta_n)/\lambda_U$ gives population feasibility, with the resulting index still below B for large n. Hence $b^\star \leq t + C(d_n + \eta_n)$. Let the optimization error tend to zero and combine the two bounds. ■

Proof of Corollary C.3. Under the null,

$$P(\varphi_n = 1) \leq P(\hat{b}_n^\star - b^\star > \lambda_n) \longrightarrow 0,$$

since $\hat{b}_n^\star - b^\star = O_p(r_n)$ and $r_n = o(\lambda_n)$. Under a fixed alternative let $\Delta = b^\star - b_0 > 0$. Eventually $\lambda_n < \Delta/2$, and $P(\varphi_n = 0) \leq P(b^\star - \hat{b}_n^\star \geq \Delta/2) \to 0$. These are pointwise-in-distribution statements; uniform size control would require uniform versions of the underlying rate assumptions. ■